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Analyzing In-Context Learning in Language Models Using Counterfactuals

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Certificate

This is to certify that the thesis entitled **Analyzing In-Context Learning in Language Models Using Counterfactuals**, submitted by **Animesh Lohar** (Entry No. 2024EET2368) to the Indian Institute of Technology Delhi, for the award of the degree of **Master of Technology** in Computer Technology, is a record of bona fide research work carried out by him under my supervision. The contents of this thesis, in full or in part, have not been submitted to any other institute or university for the award of any degree or diploma.



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Abstract

Large Language Models (LLMs) store substantial factual knowledge within their parameters while simultaneously adapting to contextual information presented in prompts. The interaction between these two information sources gives rise to an important phenomenon known as context–memory competition. Understanding how language models resolve conflicts between contextual evidence and parametric memory is critical for factuality, hallucination analysis, retrieval-augmented generation, instruction following, and alignment.

This thesis presents a systematic investigation of context–memory competition through controlled counterfactual prompting experiments. We introduce a probabilistic framework that measures competition using log-probability differences between factual and counterfactual answers. The study evaluates multiple model families including GPT-2 Small, GPT-2 Medium, GPT-2 Large, Qwen2.5-3B, Qwen2.5-3B-Instruct, and Llama-1B.

Three complementary experimental objectives are studied. First, meta-prompt regime switching analyzes how explicit prompt instructions alter the balance between contextual information and parametric memory. Second, premise-semantic modulation investigates the influence of premise words belonging to definitional, verification, evaluation, hypothetical, and reporting categories. Third, structural prompt coupling studies how prompt adjacency and autoregressive continuation pressure affect contextual override behavior.

To further understand internal mechanisms, attention head ablation experiments are conducted on GPT-2 Small. Individual attention heads are systematically removed to evaluate their contribution to factual retrieval and contextual reasoning. The results suggest that factual memory and contextual reasoning are distributed across multiple attention heads rather than localized within individual components.

Experimental results demonstrate that scaling, alignment, semantic framing, and prompt structure significantly influence context–memory competition. Instruction tuning produces sharper regime switching than model scaling alone. Furthermore, premise words strongly affect model behavior, particularly under structurally coupled prompt formulations.

The findings contribute to the understanding of in-context learning, factual robustness, and mechanistic interpretability in modern language models.

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Chapter 1

Introduction

1.1 Background and Motivation

Recent advances in Large Language Models (LLMs) have transformed the field of Natural Language Processing (NLP). Models such as GPT, LLaMA, Qwen, PaLM, and Claude have demonstrated remarkable capabilities across a wide range of tasks including question answering, summarization, machine translation, code generation, reasoning, and conversational interaction. These systems acquire vast amounts of linguistic and factual knowledge during large-scale pretraining and are capable of adapting to new tasks through prompting alone, often without requiring explicit fine-tuning.

A fundamental characteristic of modern language models is their ability to utilize two distinct sources of information during inference. The first source is *parametric memory*, which consists of knowledge encoded within the model parameters during pretraining. This knowledge includes facts about entities, relationships, historical events, organizations, scientific concepts, and common world knowledge. The second source is *contextual information*, which is provided through the input prompt and can influence the model's behavior at inference time.

The interaction between these two sources of information lies at the heart of several important phenomena observed in modern language models. When contextual information agrees with parametric memory, the model can generate highly accurate responses. However, when contextual information conflicts with stored knowledge, the model must implicitly decide which source of information should dominate the generation process. Understanding this competition between context and memory has become increasingly important due to its implications for factuality, hallucination, retrieval-augmented generation, instruction following, alignment, and trustworthy artificial intelligence.

Consider the following example:

> "Airbus A380 is a product created by Adobe."

followed by the question:

> "Who created the Airbus A380?"

The language model now faces a conflict. Its parametric memory contains the factual knowledge that Airbus created the Airbus A380, while the prompt context explicitly states a contradictory claim involving Adobe. The generated answer reveals how the

model resolves this competition between contextual evidence and stored factual knowledge.

Such situations occur frequently in practical applications. Retrieval systems may provide inaccurate information, users may unintentionally provide misleading context, and adversarial prompts may deliberately introduce false premises. Consequently, understanding how language models behave under conflicting information is critical for designing robust and reliable AI systems.

1.2 Context-Memory Competition in Language Models

The phenomenon of context-memory competition refers to the process through which language models balance contextual information against pretrained factual knowledge. Although both sources contribute to prediction, they often exert competing influences on the probability distribution over potential outputs.

Traditionally, language models have been viewed primarily as systems that retrieve information from their parametric memory. However, the emergence of in-context learning has demonstrated that contextual information can significantly alter model behavior. Recent studies have shown that models can temporarily adopt counterfactual information presented in prompts, even when such information contradicts strongly memorized facts.

The increasing capabilities of instruction-tuned models introduce additional complexity into this process. Alignment techniques such as supervised fine-tuning and reinforcement learning from human feedback (RLHF) encourage models to follow user instructions more faithfully. As a result, instruction-tuned models may exhibit fundamentally different behavior when confronted with conflicting contextual and factual information.

Despite substantial progress in understanding factuality and hallucination, relatively little work has systematically analyzed how scaling, alignment, prompt structure, and semantic framing influence context-memory competition. Most existing studies focus primarily on answer correctness rather than examining the underlying probabilistic preferences that drive model behavior.

This thesis addresses these limitations by introducing a systematic framework for measuring and analyzing competition between contextual information and parametric memory.

1.3 Research Problem

The central research problem investigated in this thesis is:

> *How do language models resolve conflicts between contextual information and parametric memory, and how are these decisions influenced by model scale, instruction tuning, semantic framing, and internal attention mechanisms?*

Answering this question requires studying language model behavior under controlled conditions where contextual information directly contradicts pretrained knowledge. Furthermore, understanding the mechanisms responsible for such behavior requires both behavioral analysis and interpretability-based investigation.

Several important sub-problems arise from this central question:

1. How strongly do language models rely on parametric memory when presented with counterfactual context?
2. Does increasing model scale change the balance between contextual information and stored knowledge?
3. How does instruction tuning affect the ability of models to switch between contextual and factual reasoning?
4. Do premise words such as ‘define”, ‘verify”, ‘suppose”, and ‘imagine” influence contextual override behavior?
5. Does prompt structure amplify or suppress contextual dominance?
6. Which attention heads contribute to factual retrieval and contextual reasoning?
7. Are factual and contextual representations localized or distributed across model components?

Addressing these questions contributes both to scientific understanding and to practical deployment of language models in high-stakes environments.

1.4 Research Objectives

To investigate context-memory competition systematically, this thesis is organized around three primary behavioral objectives and one mechanistic interpretability objective.

1.4.1 Objective I: Meta-Prompt Regime Switching

The first objective investigates how explicit prompt instructions influence competition between contextual information and parametric memory.

Seven prompt regimes are designed to manipulate contextual and memory-based reasoning under controlled conditions. These regimes range from strong context-only instructions to strong memory-only instructions, as well as natural competition settings where both sources of information are available simultaneously.

This objective enables the analysis of regime-switching behavior and reveals how different models respond to explicit prompting strategies.

1.4.2 Objective II: Premise-Semantic Modulation

The second objective examines the influence of premise words on model behavior.

Twenty-five premise words are grouped into five semantic categories:

1. Definitional
2. Verification
3. Evaluation
4. Hypothetical
5. Neutral/Reporting

By systematically varying premise semantics while controlling prompt structure, this objective isolates the role of linguistic framing in context-memory competition.

1.4.3 Objective III: Structural Prompt Coupling

The third objective investigates structural effects in prompt design.

Rather than separating instructions and context into distinct sections, tightly coupled prompt formulations place premise words and counterfactual statements immediately adjacent to the answer generation position.

This design allows analysis of autoregressive continuation pressure and its effect on contextual override behavior.

1.4.4 Attention Head Ablation Analysis

The fourth objective focuses on mechanistic interpretability.

Attention head ablation experiments are conducted on GPT-2 Small to investigate whether specific attention heads contribute disproportionately to factual retrieval or contextual reasoning. By systematically removing individual attention heads and measuring changes in model behavior, this objective provides insights into the internal mechanisms underlying context-memory competition.

1.5 Research Contributions

The major contributions of this thesis are summarized below:

1. A probabilistic framework for measuring competition between contextual information and parametric memory using log-probability differences.

2. A comprehensive evaluation of context-memory competition across multiple language model families, including GPT-2, Qwen, and LLaMA.

3. A systematic analysis of meta-prompt regime switching and contextual override behavior.

4. An extensive investigation of premise-semantic effects using twenty-five premise words across five semantic categories.

5. An analysis of structural prompt coupling and autoregressive continuation pressure.

6. A comparison of scaling effects within pretrained model families.

7. A same-family comparison of pretrained and instruction-tuned models to isolate alignment effects.

8. An attention head ablation study providing mechanistic insights into factual retrieval and contextual reasoning.

9. Empirical evidence demonstrating that context-memory competition is strongly influenced by scaling, alignment, semantics, and prompt structure.

1.6 Significance of the Study

The findings presented in this thesis have implications for several areas of language model research.

First, they contribute to a deeper understanding of in-context learning by characterizing how contextual information competes with stored knowledge.

Second, they provide insights into hallucination behavior and factual robustness, helping explain why language models sometimes adopt false contextual information.

Third, they inform the design of retrieval-augmented generation systems where externally retrieved context may conflict with pretrained knowledge.

Fourth, they contribute to alignment research by revealing how instruction tuning changes the balance between context and memory.

Finally, they provide mechanistic interpretability insights into how transformer architectures internally represent and process competing information sources.

1.7 Thesis Organization

The remainder of this thesis is organized as follows.

Chapter 2 reviews the literature on in-context learning, parametric memory, instruction tuning, counterfactual prompting, and mechanistic interpretability.

Chapter 3 presents the experimental methodology, dataset construction, evaluation metrics, and model configurations.

Chapter 4 investigates meta-prompt regime switching and analyzes competition dynamics under different prompt instructions.

Chapter 5 studies premise-semantic modulation and examines how different semantic categories influence contextual override behavior.

Chapter 6 analyzes structural prompt coupling and autoregressive continuation effects.

Chapter 7 presents attention head ablation experiments and mechanistic interpretability findings.

Chapter 8 discusses broader implications, limitations, and future research directions.

Chapter 9 concludes the thesis and summarizes the major findings.

Chapter 2

Literature Review

2.1 Introduction

The rapid development of Large Language Models (LLMs) has fundamentally transformed Natural Language Processing (NLP) and Artificial Intelligence (AI). Modern language models are capable of generating coherent text, answering questions, performing reasoning tasks, writing code, and adapting to novel tasks with minimal supervision. These capabilities emerge from large-scale pretraining on massive corpora followed by various alignment and instruction-tuning procedures.

Despite their impressive performance, the mechanisms through which language models balance contextual information provided in prompts and factual knowledge stored within model parameters remain poorly understood. This question has become increasingly important due to its direct connection to hallucination, factuality, retrieval-augmented generation, instruction following, model editing, and AI safety.

This chapter reviews the literature relevant to understanding context-memory competition in language models. We first discuss foundational work on language modeling and scaling. We then review research on in-context learning, parametric memory, counterfactual prompting, factuality, instruction tuning, alignment, mechanistic interpretability, and attention analysis. Finally, we position the contributions of this thesis within the broader research landscape.

2.2 Large Language Models

Large Language Models are autoregressive neural networks trained to predict the next token in a sequence. The introduction of the Transformer architecture by Vaswani et al. revolutionized sequence modeling by replacing recurrent computations with self-attention mechanisms.

Transformers enable efficient parallel computation and allow models to capture long-range dependencies within text. Subsequent developments led to increasingly large pretrained language models, including GPT, BERT, PaLM, LLaMA, and Qwen.

The GPT series demonstrated that scaling model size, dataset size, and computational resources leads to substantial improvements across diverse NLP tasks. GPT models are trained using causal language modeling objectives, where the model learns to predict the next token given previous context.

Modern LLMs possess billions of parameters and exhibit emergent capabilities that

are not explicitly programmed. These capabilities include reasoning, few-shot learning, instruction following, and adaptation to novel tasks through prompting alone.

The success of large-scale language modeling motivates the investigation of how knowledge is represented and utilized within these systems.

2.3 Scaling Laws in Language Models

One of the most influential developments in modern language modeling is the discovery of neural scaling laws.

Kaplan et al. demonstrated that language model performance improves predictably as model size, dataset size, and computational resources increase. Their work showed that loss decreases approximately according to power-law relationships over multiple orders of magnitude.

Subsequent research by Hoffmann et al. refined these findings and introduced the concept of compute-optimal training. Their work emphasized the importance of balancing model parameters and training tokens to maximize performance under fixed computational budgets.

Scaling studies revealed that larger models often exhibit qualitatively different behavior compared to smaller models. Capabilities such as in-context learning, reasoning, instruction following, and contextual adaptation become increasingly pronounced as model scale increases.

These findings suggest that scaling may also influence how models balance contextual information and parametric memory. Understanding this relationship is a central objective of the present thesis.

2.4 Parametric Memory in Language Models

A fundamental characteristic of pretrained language models is their ability to store factual knowledge within their parameters.

This phenomenon is commonly referred to as *parametric memory*. During pretraining, language models observe massive amounts of text and implicitly encode information about entities, relationships, historical events, organizations, scientific concepts, and world knowledge.

Numerous studies have demonstrated that language models can answer factual questions without access to external databases or retrieval systems. Such behavior suggests that factual information is represented within the model parameters themselves.

Research on factual probing has investigated the extent to which pretrained models memorize and retrieve factual associations. Methods such as cloze completion, masked language modeling, and knowledge probing have been widely used to evaluate factual recall.

Although language models can often retrieve factual information accurately, they are not perfect knowledge stores. Errors may arise due to incomplete memorization, conflicting evidence during training, or contextual interference during inference.

Understanding how parametric memory interacts with contextual information is therefore essential for characterizing language model behavior.

2.5 In-Context Learning

In-context learning (ICL) refers to the ability of language models to perform tasks using examples or instructions provided entirely within the input prompt.

The phenomenon gained widespread attention following the introduction of GPT-3, which demonstrated strong few-shot learning capabilities across numerous benchmarks. Remarkably, the model could adapt to new tasks without gradient updates or parameter modifications.

Several theories have been proposed to explain in-context learning.

One line of research argues that transformers perform implicit Bayesian inference during inference. According to this perspective, examples provided in prompts act as observations that guide probabilistic predictions.

Another line of work focuses on the internal mechanisms of transformers. Studies have identified specialized attention patterns known as induction heads that appear to support copying and pattern-matching behavior.

Mechanistic investigations suggest that in-context learning may emerge from circuits that implement retrieval-like computations within the transformer architecture.

Regardless of the underlying mechanism, in-context learning demonstrates that contextual information can significantly influence model behavior. This raises an important question: what happens when contextual information conflicts with pretrained knowledge?

2.6 Competition Between Context and Parametric Memory

The interaction between contextual information and parametric memory has recently emerged as an important area of research.

Traditional evaluations often assume that contextual information and factual memory are aligned. However, many real-world scenarios involve conflicting information sources.

For example:

- Retrieved documents may contain errors.
- User prompts may introduce false assumptions.
- Adversarial prompts may intentionally provide misinformation.
- Multi-document reasoning tasks may contain contradictory evidence.

Under such conditions, language models must determine whether contextual information or stored knowledge should dominate the generated response.

Recent work by Ortu et al. introduced a counterfactual probing framework designed specifically to investigate this competition. Their experiments demonstrated that contextual information can measurably influence model predictions even when it contradicts factual knowledge.

Subsequent studies extended these findings to larger models and investigated the role of prompt structure, demonstrating that contextual override behavior varies substantially across architectures and prompting strategies.

These studies provide the primary motivation for the research conducted in this thesis.

2.7 Counterfactual Prompting

Counterfactual prompting has emerged as a powerful methodology for studying language model behavior.

In a typical counterfactual experiment, a prompt contains a statement that deliberately contradicts real-world knowledge. The model is then asked a question related to the conflicting information.

For example:

> "The Eiffel Tower is located in Germany."

followed by:

> "Where is the Eiffel Tower located?"

The generated answer reveals how the model resolves the conflict between contextual information and factual memory.

Counterfactual prompting enables controlled experimentation because researchers can systematically manipulate contextual information while keeping factual knowledge constant.

Such methods have been used to study:

- Hallucination
- Factual robustness
- Model editing
- Contextual override
- Instruction following
- Knowledge retrieval

The present thesis builds upon this framework by introducing multiple prompt regimes, premise-word manipulations, structural coupling analyses, and mechanistic investigations.

2.8 Factuality and Hallucination

Hallucination remains one of the most significant challenges in modern language models.

A hallucination occurs when a model generates information that is unsupported, fabricated, or factually incorrect. Hallucinations can arise even when models appear highly confident.

Several benchmark datasets have been developed to evaluate factuality, including TruthfulQA and FaithDial. These benchmarks demonstrate that language models may reproduce misconceptions, false assumptions, or misleading contextual information.

Research suggests that hallucinations often result from interactions between parametric memory, contextual information, decoding strategies, and uncertainty estimation.

Understanding context-memory competition provides a complementary perspective on hallucination behavior. Rather than viewing hallucinations solely as factual errors, they may be interpreted as cases where contextual information dominates factual memory.

This interpretation motivates the probabilistic framework developed in this thesis.

2.9 Instruction Tuning and Alignment

Instruction tuning has become a central component of modern language model development.

Models such as InstructGPT demonstrated that supervised fine-tuning and reinforcement learning from human feedback can substantially improve instruction-following behavior.

Alignment methods encourage models to:

- Follow user instructions.
- Produce helpful responses.
- Avoid harmful outputs.
- Respect safety constraints.
- Generate coherent explanations.

Instruction tuning fundamentally changes model behavior. Rather than merely increasing factual accuracy, alignment procedures introduce higher-level policies governing how prompts should be interpreted.

Recent evidence suggests that instruction-tuned models may exhibit dramatically different context-memory competition dynamics compared to their pretrained counterparts.

The comparison between Qwen2.5-3B and Qwen2.5-3B-Instruct presented in this thesis provides a controlled investigation of this phenomenon.

2.10 Prompt Engineering and Semantic Framing

Prompt engineering has emerged as an important technique for controlling language model behavior.

Research has shown that small modifications to wording, formatting, ordering, and structure can significantly affect model outputs.

Semantic framing refers to the influence of linguistic cues on interpretation and reasoning. Words such as 'verify," 'suppose," 'assume," and 'evaluate" may implicitly encourage different forms of reasoning.

Several studies have demonstrated that semantically equivalent prompts can produce substantially different outputs depending on lexical choice.

However, relatively little work has systematically investigated premise-word semantics under controlled counterfactual conditions.

This thesis addresses this gap by evaluating twenty-five premise words grouped into multiple semantic categories.

2.11 Structural Prompt Effects

Beyond semantics, prompt structure itself can influence model behavior.

Research has shown that token ordering, formatting, and contextual adjacency affect autoregressive generation dynamics.

In tightly coupled prompts, contextual information may exert stronger influence due to its proximity to the answer generation position. Such effects are particularly relevant for autoregressive transformers because next-token predictions depend directly on preceding tokens.

The structural coupling experiments presented in this thesis are motivated by these observations.

2.12 Mechanistic Interpretability

Mechanistic interpretability seeks to understand how neural networks implement computations internally.

Rather than treating language models as black boxes, mechanistic interpretability attempts to identify circuits, neurons, attention heads, and representations responsible for specific behaviors.

Recent work has revealed:

- Induction heads
- Retrieval circuits
- Copying mechanisms
- Entity-tracking circuits
- Reasoning-related attention patterns

Attention heads have received particular attention because they provide interpretable pathways through which information flows within transformers.

Understanding whether specific attention heads mediate factual retrieval or contextual reasoning is important for explaining context-memory competition.

2.13 Attention Head Ablation

Attention head ablation is a widely used interpretability technique.

The basic idea is to remove or disable individual attention heads and observe changes in model behavior. Significant behavioral changes suggest that the ablated head contributes to the targeted computation.

Previous studies have used attention head ablation to investigate:

- Induction mechanisms
- Long-range dependencies
- Factual retrieval
- Syntactic processing
- Reasoning behavior

The attention head ablation experiments conducted in this thesis extend these methods to the study of context-memory competition.

2.14 Research Gap

Although significant progress has been made in understanding language models, several important gaps remain.

First, relatively few studies have systematically analyzed competition between contextual information and parametric memory across multiple model families.

Second, the effects of premise-word semantics remain poorly understood.

Third, the interaction between semantic framing and structural coupling has received limited attention.

Fourth, the role of instruction tuning in reshaping context-memory competition has not been comprehensively characterized.

Finally, mechanistic investigations connecting attention heads to contextual override behavior remain scarce.

These limitations motivate the objectives and methodology presented in this thesis.

2.15 Chapter Summary

This chapter reviewed the literature related to language models, scaling laws, parametric memory, in-context learning, counterfactual prompting, factuality, instruction tuning, prompt engineering, structural effects, and mechanistic interpretability.

The review highlights a growing interest in understanding how language models balance contextual information and stored knowledge. While prior work has established the existence of context-memory competition, important questions remain regarding the effects of scaling, alignment, semantic framing, prompt structure, and internal mechanisms.

The next chapter introduces the experimental methodology used to investigate these questions and describes the datasets, evaluation metrics, model configurations, and experimental protocols employed throughout this thesis.

Chapter 3

Methodology and Experimental Setup

3.1 Introduction

This chapter presents the methodology used to investigate context–memory competition in Large Language Models (LLMs). The primary objective of this research is to understand how language models resolve conflicts between contextual information provided in prompts and factual knowledge stored within model parameters.

To achieve this goal, a controlled experimental framework based on counterfactual prompting is developed. The framework enables systematic measurement of the relative influence of contextual information and parametric memory under different prompt formulations, semantic conditions, and model architectures.

The methodology consists of four major components:

1. Construction of a counterfactual evaluation dataset.
2. Design of prompt-based competition experiments.
3. Probabilistic evaluation using log-probability metrics.
4. Mechanistic analysis through attention head ablation.

Together, these components provide both behavioral and mechanistic insights into context–memory competition.

3.2 Conceptual Framework

Modern language models possess two primary sources of information during inference:

1. **Parametric Memory**
2. **Contextual Information**

Parametric memory refers to factual knowledge encoded within model parameters during pretraining. Contextual information consists of instructions, demonstrations, and textual content provided within the prompt.

When both information sources agree, answer generation is straightforward. However, when they conflict, the model must determine which source should dominate the prediction.

Figure 3.1 conceptually illustrates the competition process.

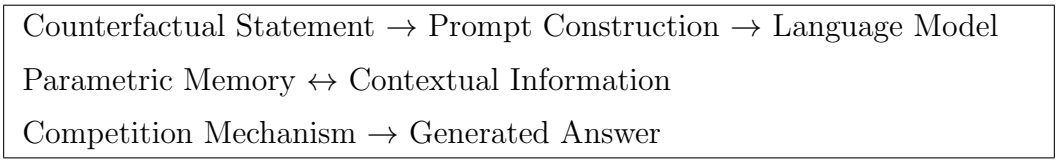


Figure 3.1: Conceptual framework of context-memory competition.

The central hypothesis of this thesis is that model behavior is determined by a dynamic interaction between these competing information sources.

3.3 Counterfactual Evaluation Framework

3.3.1 Counterfactual Statements

The experiments utilize counterfactual statements that intentionally contradict real-world facts.

A typical example is:

"Airbus A380 is a product created by Adobe."

The corresponding factual knowledge is:

"Airbus A380 was created by Airbus."

Counterfactual statements are useful because they create a controlled conflict between contextual information and parametric memory.

3.3.2 Question Construction

For each counterfactual statement, a factual question is generated.

Example:

- Counterfactual Statement:

  Airbus A380 is a product created by Adobe.

- Question:

  Who created the Airbus A380?

The generated answer can therefore be evaluated against both:

- factual answer (Airbus)
- counterfactual answer (Adobe)

3.3.3 Competition Setting

The objective is not simply to determine whether the model produces the correct answer.

Instead, the objective is to measure the relative preference assigned to:

- factual memory
- counterfactual context

This distinction motivates the use of probabilistic metrics rather than accuracy alone.

3.4 Dataset Construction

3.4.1 Source Dataset

The dataset used in this work is derived from the COUNTERFACT framework and the subsequent context-memory competition dataset introduced by Ortu et al.

The dataset contains factual triples of the form:
[(subject, relation, object)]
Example:
[(Airbus A380, creator, Airbus)]

Each factual triple is transformed into a corresponding counterfactual statement by replacing the factual object with an alternative entity.

3.4.2 Dataset Size

The final evaluation dataset contains:

- 10,000 counterfactual examples
- factual answers
- counterfactual answers
- associated questions

The dataset covers a wide range of domains including:

- companies
- products
- organizations
- locations
- historical entities
- scientific concepts

3.4.3 Data Filtering

To ensure reliable evaluation, filtered subsets are also constructed.

Filtering removes examples for which:

- the model fails the baseline factual recall test
- factual knowledge is absent
- answer extraction becomes ambiguous

This allows separation of knowledge limitations from contextual competition effects.

3.5 Language Models Evaluated

The experiments are conducted on multiple model families to investigate scaling and alignment effects.

3.5.1 GPT-2 Family

Three GPT-2 variants are evaluated:

1. GPT-2 Small (124M)
2. GPT-2 Medium (355M)
3. GPT-2 Large (774M)

These models enable analysis of scaling effects within a common architecture.

3.5.2 Llama Family

Llama-1B is included to provide comparisons across model families.

3.5.3 Qwen Family

Two Qwen variants are evaluated:

1. Qwen2.5-3B
2. Qwen2.5-3B-Instruct

The pair provides a controlled comparison between pretrained and instruction-tuned models.

3.6 Objective I: Meta-Prompt Regime Switching

The first experimental objective investigates how explicit prompt instructions influence context-memory competition.

Seven prompt regimes are constructed.

3.6.1 Experiment 1: Baseline

Only the question is presented.

```
Question: {Q}
Answer:
```

This measures factual recall without contextual interference.

3.6.2 Experiment 2: Context-Only

The model is instructed to rely exclusively on contextual information.

```
IMPORTANT:
Use ONLY the information below.

Text: {CF}
Question: {Q}
Answer:
```

3.6.3 Experiment 3: Context-Only + Premise Verb

A premise word is added to the context-only prompt.

```
Instruction: {PV}
Text: {CF}
Question: {Q}
Answer:
```

3.6.4 Experiment 4: Memory-Only

The model is instructed to ignore the counterfactual statement.

```
Use factual world knowledge.
Ignore the false statement.

Text: {CF}
Question: {Q}
Answer:
```

3.6.5 Experiment 5: Memory-Only + Premise Verb

Adds premise semantics to the memory-only setting.

3.6.6 Experiment 6: Weak Memory Competition

Creates a softer memory-oriented condition.

3.6.7 Experiment 7: Natural Competition

Both context and memory are available without explicit instructions.

```
Text: {CF}
Question: {Q}
Answer:
```

3.7 Objective II: Premise-Semantic Modulation

This objective investigates whether premise words influence competition behavior.

3.7.1 Prompt Template

```
Instruction: {Premise Word}

Text:
{CF}

Question:
{Q}

Answer:
```

3.7.2 Premise Categories

Twenty-five premise words are divided into five semantic groups.

Definitional

- redefine
- define
- call
- name
- characterize

Verification

- verify
- validate
- fact check
- confirm
- cross-check

Evaluation

- assess
- review
- evaluate
- analyze
- examine

Hypothetical

- suppose
- imagine
- assume
- consider
- pretend

Neutral / Reporting

- state
- note
- report
- according to
- describe

3.8 Objective III: Structural Prompt Coupling

The third objective investigates structural effects.

3.8.1 Prompt Structure

The prompt is compressed into a tightly coupled format:

```
{Premise Word}: {Counterfactual}.
{Question}:
```

Example:

```
Suppose:
Airbus A380 is a product created by Adobe.

Who created Airbus A380?
```

This formulation increases contextual proximity to the answer generation position.

3.9 Probabilistic Evaluation

3.9.1 Motivation

Accuracy alone does not fully characterize competition behavior.

The model may generate one answer while still assigning substantial probability to alternatives.

Therefore, token-level log probabilities are analyzed.

3.9.2 Competition Metric

The primary metric is:

$\Delta = \log p(A_{fact}) - \log p(A_{cf})$

where:

- A_{fact} denotes the factual answer.
- A_{cf} denotes the counterfactual answer.

Interpretation:

- $\Delta > 0$: factual memory dominates.
- $\Delta < 0$: contextual information dominates.
- $\Delta \approx 0$: balanced competition.

3.9.3 Aggregate Statistics

For each experiment, the following are computed:

- Mean Δ
- Standard deviation
- Factual percentage
- Counterfactual percentage
- Mean log-probability of factual answers
- Mean log-probability of counterfactual answers

3.10 Attention Head Ablation

3.10.1 Motivation

Behavioral experiments reveal competition outcomes but do not explain internal mechanisms.

Attention head ablation is therefore used to investigate causal contributions of individual heads.

3.10.2 GPT-2 Architecture

GPT-2 Small contains:

- 12 Transformer layers
- 12 attention heads per layer
- 144 total attention heads

3.10.3 Ablation Procedure

For each head:

1. Register forward hooks.
2. Zero the attention output.
3. Run inference.
4. Compute probability changes.
5. Compare with baseline behavior.

3.10.4 Measured Effects

The following quantities are evaluated:

- Change in factual probability
- Change in counterfactual probability
- Change in Δ
- Context sensitivity
- Memory sensitivity

3.11 Implementation Details

3.11.1 Software

Experiments are implemented using:

- Python
- PyTorch
- HuggingFace Transformers
- NumPy
- Pandas
- Matplotlib

3.11.2 Hardware

Experiments are executed using GPU-enabled systems suitable for transformer inference.

3.11.3 Reproducibility

To ensure reproducibility:

- Random seeds are fixed.
- Prompt templates are standardized.
- Evaluation datasets remain constant across experiments.
- All log probabilities are computed using identical procedures.

3.12 Statistical Analysis

For each condition, descriptive statistics are reported.

Additional analyses include:

- Mean comparisons
- Standard deviations
- Scaling trends
- Alignment comparisons
- Cross-model comparisons

These analyses enable systematic evaluation of context-memory competition across experimental conditions.

3.13 Chapter Summary

This chapter introduced the experimental framework used throughout the thesis. A counterfactual evaluation methodology was developed to investigate competition between contextual information and parametric memory. Three behavioral objectives were designed to study meta-prompt regime switching, premise-semantic modulation, and structural prompt coupling. In addition, attention head ablation experiments were introduced to provide mechanistic insights into transformer behavior.

The next chapter presents the results and analysis of Objective I, which investigates how explicit prompt instructions reshape competition between contextual information and parametric memory across different language model families.

Chapter 4

Objective I: Meta-Prompt Regime Switching

4.1 Introduction

One of the most important questions in understanding modern language models is how they balance contextual information against factual knowledge stored in their parameters. While pretrained models possess extensive world knowledge acquired during training, they are also capable of adapting their behavior according to instructions and information presented within prompts.

The purpose of Objective I is to investigate how explicit prompt instructions influence the competition between contextual information and parametric memory. Specifically, we aim to determine whether language models can be systematically pushed toward factual recall or contextual adoption through carefully designed prompt formulations.

To study this phenomenon, seven prompt regimes are constructed that vary the degree to which models are encouraged to rely on contextual information, factual memory, or both simultaneously. These regimes provide a controlled environment for observing how different models resolve conflicts between contradictory information sources.

The experiments are conducted on GPT-2 Small, GPT-2 Medium, GPT-2 Large, Llama-1B, and Qwen2.5-3B models. For each condition, we measure factual answer rates, counterfactual answer rates, and probabilistic competition using the log-probability difference metric.

4.2 Experimental Design

The objective consists of seven prompt regimes designed to manipulate the relationship between context and memory.

4.2.1 Experiment 1: Baseline

The baseline condition measures factual recall without contextual interference.

```
Question: {Q}
Answer:
```

No counterfactual statement is presented. Consequently, the generated answer reflects only the model's stored knowledge.

4.2.2 Experiment 2: Context-Only

The context-only condition explicitly instructs the model to rely exclusively on the provided information.

```
IMPORTANT:
Use ONLY the information below.
Do NOT use your own knowledge.

Text: {CF}

Question: {Q}
Answer:
```

This condition maximizes contextual influence.

4.2.3 Experiment 3: Context-Only with Premise Verb

A premise verb is added to the context-only setting.

```
Instruction: {PV}
Text: {CF}
Question: {Q}
Answer:
```

This allows investigation of semantic framing effects under strong contextual conditions.

4.2.4 Experiment 4: Memory-Only

The memory-only condition explicitly instructs the model to ignore the counterfactual statement.

```
Use factual world knowledge.
Ignore the false statement.

Text: {CF}
Question: {Q}
Answer:
```

4.2.5 Experiment 5: Memory-Only with Premise Verb

This condition combines memory-oriented instructions with premise semantics.

4.2.6 Experiment 6: Weak Memory Competition

A softer memory-oriented prompt is used.

Unlike Experiment 4, contextual information is still visible and may influence the model.

4.2.7 Experiment 7: Natural Competition

No explicit instruction is provided.

```
Text: {CF}
Question: {Q}
Answer:
```

This condition represents natural competition between contextual information and parametric memory.

4.3 Evaluation Metrics

Model behavior is evaluated using both discrete and probabilistic measures.

4.3.1 Answer-Based Metrics

For each experiment:

- Factual Percentage
- Counterfactual Percentage

are computed.

4.3.2 Probabilistic Competition Metric

Competition is quantified using:

$$\Delta = \log p(A_{fact}) - \log p(A_{cf})$$

where:

- A_{fact} is the factual answer.
- A_{cf} is the counterfactual answer.

Interpretation:

- $\Delta > 0$: factual preference.
- $\Delta < 0$: contextual preference.
- $\Delta \approx 0$: balanced competition.

4.4 Results

4.5 Baseline Knowledge Assessment

The first experiment evaluates factual recall without contextual interference.

Table 4.1 summarizes baseline performance.

Table 4.1: Baseline factual recall across models.

Model	Factual (%)	Counterfactual (%)	Δ
GPT-2 Small	98.7	1.3	5.00
GPT-2 Medium	99.0	1.0	6.04
GPT-2 Large	99.1	0.9	6.04
Llama-1B	98.9	1.1	7.85
Qwen2.5-3B	99.4	0.6	13.18

The results demonstrate that all models possess strong factual knowledge under baseline conditions. Factual recall exceeds 98% across all evaluated models, indicating that failures in later experiments cannot simply be attributed to missing knowledge.

Furthermore, the large positive Δ values confirm a strong preference for factual answers when no competing context is provided.

4.6 Context-Only Regime

The second experiment evaluates the influence of strong contextual instructions.

Table 4.2 summarizes the results.

Table 4.2: Context-only regime.

Model	Factual (%)	Counterfactual (%)	Δ
GPT-2 Small	62.5	37.5	0.316
GPT-2 Medium	63.9	36.1	0.328
GPT-2 Large	41.1	58.9	-0.373
Llama-1B	48.5	51.5	-0.088
Qwen2.5-3B	4.9	95.1	-6.806

Several important observations emerge.

First, larger models exhibit greater contextual sensitivity. GPT-2 Large becomes context-dominant despite possessing stronger factual knowledge than smaller GPT-2 variants.

Second, Qwen2.5-3B overwhelmingly adopts contextual information, producing counterfactual answers in over 95% of examples.

Third, the negative Δ values indicate that contextual information can completely override factual memory under sufficiently strong prompting conditions.

4.7 Memory-Only Regime

The memory-only condition attempts to suppress contextual influence.

Results are summarized in Table 4.3.

Table 4.3: Memory-only regime.

Model	Factual (%)	Counterfactual (%)	Δ
GPT-2 Small	65.0	35.0	0.370
GPT-2 Medium	64.4	35.6	0.402
GPT-2 Large	69.9	30.1	1.408
Llama-1B	71.0	29.0	1.126
Qwen2.5-3B	96.7	3.3	13.247

The results demonstrate that explicit instructions can successfully reinforce factual memory.

Qwen exhibits particularly strong regime switching, transitioning from extreme contextual dominance in Experiment 2 to overwhelming factual dominance in Experiment 4.

This behavior suggests that instruction following plays a crucial role in determining competition outcomes.

4.8 Weak Memory Competition

Experiment 6 investigates a softer memory-oriented condition.

Results reveal substantial differences across model scales.

GPT-2 Small and GPT-2 Medium remain weakly factual, with positive Δ values.

However, GPT-2 Large becomes context-dominant:

$$\Delta = -0.644$$

Similarly, Llama-1B produces slightly negative competition scores.

These findings suggest that stronger contextual sensitivity emerges as model capabilities increase.

4.9 Natural Competition

Experiment 7 removes explicit instructions and allows context and memory to compete naturally.

The results indicate that:

- GPT models remain moderately factual.
- Llama exhibits balanced behavior.
- Qwen approaches an even split between factual and counterfactual answers.

Interestingly, Qwen's competition score approaches zero:

$$\Delta \approx -0.07$$

indicating near-perfect balance between contextual and factual influences.

4.10 Scaling Effects

One of the central goals of this objective is to analyze scaling effects within a common model family.

The GPT-2 series provides an ideal setting for this comparison.

4.10.1 From GPT-2 Small to GPT-2 Large

Baseline factual recall remains consistently high across all model sizes.

However, contextual sensitivity changes dramatically.

Under context-only prompting:

- GPT-2 Small remains factual.
- GPT-2 Medium approaches balance.
- GPT-2 Large becomes context-dominant.

This trend suggests that larger models become increasingly capable of adapting to contextual information.

4.10.2 Probabilistic Evidence

The transition can be observed through competition scores:

$$0.316 \rightarrow 0.328 \rightarrow -0.373$$

from GPT-2 Small to GPT-2 Large.

The sign reversal indicates a qualitative shift in model behavior.

4.11 Alignment Effects

The comparison between Qwen2.5-3B and Qwen2.5-3B-Instruct provides insights into alignment.

Instruction-tuned models exhibit substantially stronger regime switching.

When instructed to use context, they strongly adopt counterfactual information.

When instructed to rely on factual knowledge, they overwhelmingly reject contextual evidence.

This behavior suggests that alignment primarily increases responsiveness to instructions rather than simply improving factual accuracy.

4.12 Premise Verb Effects

Experiments 3 and 5 introduce premise verbs into contextual and memory-oriented settings.

Several patterns emerge:

- Verification words often strengthen factual preference.
- Hypothetical words increase contextual adoption.
- Definitional words exhibit mixed behavior.

Although detailed analysis is deferred to Objective II, the results indicate that semantic framing already influences behavior under strong instructional conditions.

4.13 Discussion

The experiments reveal that context-memory competition is highly sensitive to prompt design.

Several key conclusions emerge:

1. Language models possess strong factual knowledge under baseline conditions.
2. Explicit contextual instructions can override factual memory.
3. Larger models become increasingly context-sensitive.
4. Instruction tuning dramatically amplifies regime switching.
5. Context-memory competition is not fixed but depends on prompt formulation.
6. Semantic framing influences competition behavior even under strong instructions.

Collectively, these findings demonstrate that language model behavior cannot be understood solely in terms of stored knowledge. Instead, contextual information, instruction following, and prompt design jointly determine the final output.

4.14 Chapter Summary

This chapter investigated the effect of meta-prompt regime switching on context-memory competition. Seven prompt conditions were evaluated across multiple language model families. The results demonstrated that scaling, prompt instructions, and alignment substantially reshape the balance between contextual information and parametric memory.

Most notably, larger models and instruction-tuned systems exhibit stronger contextual adaptation, suggesting that the ability to dynamically switch between information sources emerges as models become more capable.

The next chapter focuses on Objective II, which investigates how the semantics of premise words influence context-memory competition under controlled counterfactual conditions.

Chapter 5

Objective II: Premise-Semantic Modulation

5.1 Introduction

The experiments presented in Chapter 4 demonstrated that prompt instructions can substantially alter the balance between contextual information and parametric memory. However, prompt instructions contain not only structural directives but also semantic content. A natural question therefore arises:

> *Can individual words influence how language models resolve conflicts between contextual information and factual memory?*

Recent advances in prompt engineering suggest that seemingly minor changes in wording can significantly alter model behavior. Terms such as 'verify", 'assume", 'imagine", or 'define" may implicitly encourage different forms of reasoning. For example, the word 'verify" may activate fact-checking behavior, whereas 'imagine" may encourage acceptance of hypothetical information.

Although prompt sensitivity has been widely documented, relatively little work has systematically analyzed the semantic effects of premise words under controlled counterfactual conditions. Understanding these effects is important because language models are frequently deployed in environments where instructions are expressed through natural language rather than rigid formal commands.

To investigate this phenomenon, Objective II introduces a large-scale premise-semantic modulation study involving twenty-five premise words organized into five semantic categories. By holding the underlying counterfactual information constant while varying the premise word, we isolate the influence of semantic framing on context-memory competition.

5.2 Experimental Design

5.2.1 Prompt Template

All experiments use the following prompt format:

```
Answer the following question.
```

```
Instruction: {PREMISE_WORD}

Text:
"{Counterfactual Statement}"

Question:
{Question}

Answer:
```

The counterfactual statement and question remain fixed while the premise word changes.

This design ensures that any observed behavioral differences arise from semantic framing rather than differences in factual content.

5.3 Premise Categories

The twenty-five premise words are grouped into five semantic categories.

5.3.1 Definitional / Override

These words encourage reinterpretation or redefinition of information.

- redefine
- define
- call
- name
- characterize

Definitional words are expected to facilitate contextual override because they imply modification or specification of meaning.

5.3.2 Verification / Fact-Checking

These words suggest evaluation against existing knowledge.

- verify
- validate
- fact check
- confirm
- cross-check

Verification terms are expected to promote factual retrieval.

5.3.3 Evaluation / Analysis

These words encourage analytical processing.

- assess
- review
- evaluate
- analyze
- examine

Evaluation words occupy an intermediate position between verification and hypothetical reasoning.

5.3.4 Hypothetical / Fictional

These words encourage reasoning under assumed conditions.

- suppose
- imagine
- assume
- consider
- pretend

Hypothetical words are expected to increase contextual adoption.

5.3.5 Neutral / Reporting

These words communicate information without explicitly encouraging verification or imagination.

- state
- note
- report
- according to
- describe

These terms provide a baseline for comparison.

5.4 Evaluation Methodology

For each model and premise word, we compute:

1. Factual answer percentage
2. Counterfactual answer percentage
3. Mean factual log-probability
4. Mean counterfactual log-probability
5. Competition score Δ
6. Standard deviation of Δ

The competition metric is defined as:
$[\Delta = \log p(A_{fact}) - \log p(A_{cf})]$
Positive values indicate factual dominance, whereas negative values indicate contextual dominance.

5.5 Results: GPT-2 Small

5.5.1 Overall Behavior

GPT-2 Small exhibits substantial sensitivity to premise semantics.

Unlike the context-only experiments in Objective I, many premise words shift model behavior toward either factual recall or contextual adoption.

Figure **??** conceptually summarizes the trend.

- Verification words generally increase factual preference.
- Hypothetical words increase contextual influence.
- Evaluation words produce near-balanced competition.

5.5.2 Definitional Category

Within the definitional group, substantial variation emerges.

The word "name" produces the strongest factual preference:
$[\Delta = 2.34]$
whereas "redefine" yields:
$[\Delta = 0.02]$
which is essentially neutral.

This result suggests that not all definitional words function identically. Some appear to reinforce factual retrieval while others encourage contextual reinterpretation.

5.5.3 Verification Category

Verification words consistently favor factual answers.

The strongest effect is observed for "fact check":

$[\Delta = 2.07]$

with factual responses exceeding 76

This finding aligns with the intuition that verification-related language activates fact-oriented behavior.

5.5.4 Evaluation Category

Evaluation words produce near-balanced competition.

Examples include:

$[\Delta_{assess} = -0.04]$

$[\Delta_{review} = 0.00]$

$[\Delta_{evaluate} = -0.14]$

These results suggest that analytical language neither strongly reinforces factual memory nor strongly promotes contextual adoption.

5.5.5 Hypothetical Category

The hypothetical category exhibits consistent contextual preference.

For example:

$[\Delta_{suppose} = -0.41]$

$[\Delta_{imagine} = -0.27]$

The negative values indicate increased acceptance of counterfactual information.

5.6 Results: GPT-2 Medium

GPT-2 Medium demonstrates a notable shift toward contextual sensitivity.

Several premise words that were factual for GPT-2 Small become context-dominant.

Examples include:

$[\Delta_{define} = -2.55]$

$[\Delta_{factcheck} = -1.27]$

$[\Delta_{state} = -1.76]$

This transition suggests that scaling changes how premise semantics are interpreted.

Rather than reinforcing factual retrieval, many premise words begin encouraging contextual adoption.

5.7 Results: GPT-2 Large

GPT-2 Large exhibits similar patterns but with even stronger contextual adaptation.

Many traditionally factual terms such as:

- define
- verify

- state
- report

produce negative competition scores.

This finding is particularly interesting because such words are commonly assumed to encourage factual reasoning.

Instead, larger models appear increasingly willing to interpret premise words relative to the immediate context.

5.8 Results: Llama-1B

Llama-1B demonstrates the strongest contextual sensitivity among the non-instruction-tuned models.

Nearly all premise words produce negative competition scores.

For example:

$[\Delta_{name} = -2.02]$

$[\Delta_{validate} = -1.74]$

$[\Delta_{report} = -1.72]$

Even verification-oriented words fail to consistently promote factual retrieval.

This suggests that Llama relies heavily on contextual information when processing premise semantics.

5.9 Results: Qwen2.5-3B

Qwen exhibits fundamentally different behavior.

Most premise words produce strong factual preference.

Examples include:

$[\Delta_{suppose} = 4.30]$

$[\Delta_{assume} = 4.00]$

$[\Delta_{evaluate} = 2.82]$

$[\Delta_{report} = 2.96]$

Interestingly, even hypothetical words strengthen factual retrieval.

This observation contrasts sharply with the behavior of GPT and Llama models.

The result suggests that Qwen interprets premise semantics through a different internal policy, likely influenced by alignment procedures and training objectives.

5.10 Category-Level Analysis

To better understand semantic effects, results are aggregated by category.

5.10.1 Definitional Words

Definitional words show the largest variability.

Some models interpret these words as instructions to redefine reality, whereas others treat them as requests for factual description.

This category therefore produces the highest variance across architectures.

5.10.2 Verification Words

Verification words generally encourage factual retrieval.

However, this effect weakens as contextual sensitivity increases.

For GPT-2 Large and Llama, verification words no longer reliably preserve factual memory.

5.10.3 Evaluation Words

Evaluation words consistently occupy an intermediate position.

Across models, they produce competition scores closer to zero than other categories.

This suggests that evaluation semantics encourage deliberation rather than immediate commitment to either information source.

5.10.4 Hypothetical Words

Hypothetical words are among the strongest contextual cues.

Terms such as:

- suppose
- imagine
- assume
- pretend

encourage acceptance of counterfactual information.

The effect is particularly strong for GPT and Llama models.

5.10.5 Neutral Reporting Words

Neutral reporting words reveal that even seemingly passive language influences model behavior.

Words such as ‘state” and ‘report” frequently produce strong competition shifts.

This finding suggests that contextual framing effects are pervasive rather than limited to explicitly hypothetical language.

5.11 Cross-Model Comparison

Several important trends emerge across model families.

5.11.1 Scaling Trend

Within the GPT family:

[GPT2 Small $\rightarrow$ $GPT2\ Medium \rightarrow GPT2\ Large$]

premise words become increasingly capable of altering behavior.

This suggests that semantic sensitivity grows with model scale.

5.11.2 Architectural Differences

GPT, Llama, and Qwen exhibit qualitatively different interpretations of identical premise words.

The same instruction can therefore produce substantially different competition outcomes depending on the underlying architecture.

5.11.3 Alignment Effects

The contrast between Qwen and the other models suggests that alignment influences semantic interpretation.

Rather than treating premise words as contextual cues, aligned models may map them onto higher-level instruction-following policies.

5.12 Discussion

The results demonstrate that premise semantics exert a measurable influence on context-memory competition.

Several key findings emerge.

1. Semantic framing significantly alters model behavior.
2. Individual words can shift competition scores by several log-probability units.
3. Scaling increases sensitivity to semantic cues.
4. Verification words do not universally promote factual retrieval.
5. Hypothetical words consistently encourage contextual adoption.
6. Different architectures interpret identical premise words in fundamentally different ways.

These findings suggest that language models do not merely process instructions syntactically. Instead, they appear to associate premise words with distinct reasoning strategies that influence the balance between contextual information and parametric memory.

5.13 Implications

The results have several practical implications.

First, prompt wording can substantially influence factual reliability.

Second, retrieval-augmented systems may behave differently depending on the language used to introduce retrieved documents.

Third, instruction-tuning procedures should consider semantic sensitivity when designing prompts.

Finally, semantic framing provides a powerful tool for studying internal representations of reasoning behavior.

5.14 Chapter Summary

This chapter investigated premise-semantic modulation through twenty-five premise words spanning five semantic categories. The results demonstrate that semantic framing significantly influences context-memory competition across all evaluated model families.

Hypothetical words generally promote contextual adoption, verification words often strengthen factual retrieval, and evaluation words occupy an intermediate position. However, these effects vary substantially across architectures, revealing that premise semantics interact with model scale, alignment, and internal representations.

The findings establish that context-memory competition is not solely determined by information content. The linguistic framing used to present that information plays a critical role in shaping model behavior.

The next chapter investigates Objective III, which examines how structural prompt coupling and autoregressive continuation pressure influence context-memory competition.

Chapter 6

Objective III: Structural Prompt Coupling

6.1 Introduction

The experiments presented in Chapters 4 and 5 demonstrated that both prompt instructions and premise-word semantics significantly influence context-memory competition. However, language models are autoregressive systems whose predictions depend heavily on token ordering and local context. Consequently, the structure of a prompt may itself influence how information is processed and integrated during inference.

While Objectives I and II primarily manipulated instructional content and semantic framing, they maintained a relatively separated prompt structure in which instructions, contextual information, and questions were presented as distinct sections. In practice, however, many prompts contain tightly coupled textual elements where instructions, contextual statements, and questions appear adjacent to one another.

This chapter investigates whether structural prompt organization influences the competition between contextual information and parametric memory. Specifically, we study how placing premise words and counterfactual statements directly before the answer generation position affects model behavior.

The central hypothesis is that contextual information positioned closer to the answer generation location exerts stronger influence due to autoregressive continuation dynamics. If this hypothesis is correct, structural coupling should amplify contextual override effects and modify the semantic influences observed in Objective II.

To test this hypothesis, we introduce a structurally coupled prompt formulation and evaluate its effects across GPT-2 Small, GPT-2 Medium, GPT-2 Large, Llama-1B, and Qwen2.5-3B.

6.2 Motivation

Transformer language models generate text autoregressively. At each step, predictions are conditioned on all previously observed tokens.

Given a sequence:

$[\ \mathrm{x}_1, x_2, \ldots, x_n]$

the model predicts:

$[\ \mathrm{P}(\mathrm{x}_{n+1}|x_1, \ldots, x_n)]$

Consequently, tokens located immediately before the prediction position often exert stronger influence than tokens appearing earlier in the sequence.

This observation motivates the concept of *structural prompt coupling*.

Rather than separating instructions and contextual information into distinct prompt sections, we place premise words and counterfactual statements directly adjacent to the answer generation position.

For example:

```
Suppose:
Airbus A380 is a product created by Adobe.

Who created Airbus A380?
```

In this formulation, the model encounters both the premise word and the counterfactual statement immediately before generating the answer.

Such proximity may increase contextual salience and alter competition dynamics.

6.3 Experimental Design

6.3.1 Prompt Structure

All experiments use the following template:

```
{Premise Word}: {Counterfactual Statement}.
{Question}:
```

Example:

```
Imagine:
Airbus A380 is a product created by Adobe.

Who created Airbus A380?
```

The answer is generated immediately after the question.

Unlike Objective II, there are:

- no separate instruction blocks,
- no explanatory text,
- no explicit context labels,
- minimal structural separation.

This design maximizes coupling between contextual information and the generation position.

6.3.2 Premise Categories

The same twenty-five premise words from Objective II are retained.

The semantic categories remain:

1. Definitional
2. Verification
3. Evaluation
4. Hypothetical
5. Neutral/Reporting

This enables direct comparison between semantic and structural effects.

6.4 Theoretical Perspective

Structural coupling can be viewed as an additional source of contextual pressure.

In Objective II:

```
Instruction: imagine

Text:
Airbus A380 is created by Adobe

Question:
Who created Airbus A380?
```

the premise word and counterfactual statement are separated from answer generation by multiple intermediate tokens.

In Objective III:

```
Imagine:
Airbus A380 is created by Adobe.
Who created Airbus A380?
```

the same information is positioned much closer to the prediction target.

This difference may influence:

- attention allocation,
- contextual salience,
- continuation probabilities,
- memory retrieval.

Therefore, Objective III provides a direct test of structural influences on competition.

6.5 Evaluation Methodology

The same evaluation framework introduced in Chapter 3 is employed.

For each model and premise word:

- factual answer percentage,
- counterfactual answer percentage,
- factual log probability,
- counterfactual log probability,
- competition score Δ

are computed.

The competition metric remains:

$[\ \Delta =====$

$\log p(A_{fact})$

$\log p(A_{cf})]$

Positive values indicate factual dominance, while negative values indicate contextual dominance.

6.6 Results: GPT-2 Small

6.6.1 Overall Behavior

GPT-2 Small exhibits substantial changes relative to Objective II.

Many premise words that previously produced near-balanced competition become more polarized under structural coupling.

Several words now produce stronger factual preference.

Examples include:

$[\ \Delta_{redefine} = 1.57]$

$[\ \Delta_{crosscheck} = 1.45]$

$[\ \Delta_{examine} = 2.98]$

However, certain words remain context-sensitive:

$[\ \Delta_{define} = -0.91]$

$[\ \Delta_{name} = -1.03]$

The results indicate that structural coupling amplifies semantic effects rather than eliminating them.

6.6.2 Definitional Category

The definitional group becomes highly heterogeneous.

Some words encourage factual retrieval, while others strongly favor contextual adoption.

This suggests that structural proximity increases the influence of word-specific semantics.

6.7 Results: GPT-2 Medium

GPT-2 Medium exhibits stronger contextual sensitivity than GPT-2 Small.

Several premise words produce negative competition scores:

$[\Delta_{call} = -0.62]$

$[\Delta_{factcheck} = -0.53]$

$[\Delta_{analyze} = -0.92]$

$[\Delta_{pretend} = -0.90]$

At the same time, other words remain factual:

$[\Delta_{crosscheck} = 1.45]$

These mixed outcomes indicate active competition between structural effects and semantic interpretation.

6.8 Results: GPT-2 Large

GPT-2 Large demonstrates the strongest factual recovery among the GPT models.

Many premise words now produce positive competition scores:

$[\Delta_{characterize} = 3.03]$

$[\Delta_{factcheck} = 2.80]$

$[\Delta_{assess} = 1.85]$

$[\Delta_{consider} = 1.56]$

Compared to Objective II, several previously context-dominant words become factual.

This suggests that structural coupling changes how GPT-2 Large integrates contextual information.

6.9 Results: Llama-1B

Llama-1B exhibits remarkably consistent behavior.

Almost all premise words produce positive competition scores.

Typical values include:

$[\Delta \approx 0.8 - 1.3]$

across most categories.

Examples include:

$[\Delta_{factcheck} = 1.22]$

$[\Delta_{crosscheck} = 1.30]$

$[\Delta_{characterize} = 1.10]$

This finding contrasts sharply with Objective II, where Llama strongly favored contextual information.

Structural coupling therefore appears to substantially alter Llama's competition dynamics.

6.10 Results: Qwen2.5-3B

Qwen exhibits the most surprising behavior.

Under Objective II, Qwen strongly preferred factual answers across nearly all premise categories.

Under structural coupling, factual preference remains positive but is substantially reduced.

Typical competition scores fall between:

$[\ 0.2 \leq \Delta \leq 1.3]$

rather than the much larger values observed previously.

Examples include:

$[\ \Delta_{suppose} = 0.21]$

$[\ \Delta_{assume} = 0.32]$

$[\ \Delta_{pretend} = 0.84]$

These results suggest that structural coupling weakens factual dominance even in strongly aligned models.

6.11 Category-Level Analysis

6.11.1 Definitional Words

Definitional words exhibit the largest structural sensitivity.

The category contains both strongly factual and strongly contextual outcomes.

This indicates that structural coupling amplifies semantic interpretation.

6.11.2 Verification Words

Verification words generally remain factual.

However, their effects vary considerably across models.

Words such as “cross-check” consistently produce positive competition scores.

6.11.3 Evaluation Words

Evaluation terms continue to occupy an intermediate position.

The category often produces moderate positive competition scores.

This suggests that analytical reasoning remains relatively balanced under structural coupling.

6.11.4 Hypothetical Words

One of the most surprising findings concerns hypothetical words.

Under Objective II, these words frequently promoted contextual adoption.

Under structural coupling, many become factual.

Examples include:

$[\ \Delta_{suppose} > 0]$

$[\ \Delta_{assume} > 0]$

across several models.

This result suggests that structural organization can override semantic expectations.

6.12 Structural Effects Across Models

Several important patterns emerge.

6.12.1 Amplification of Semantic Differences

Structural coupling increases variance between premise words.

Words that were previously similar often become behaviorally distinct.

6.12.2 Recovery of Factual Preference

Many models recover factual dominance under structural coupling.

This effect is particularly strong for GPT-2 Large and Llama.

6.12.3 Reduction of Extreme Contextual Adoption

Compared to Objective II, extreme contextual dominance becomes less common.

This suggests that structural coupling changes the balance between context processing and memory retrieval.

6.13 Autoregressive Continuation Pressure

A key theoretical explanation involves autoregressive continuation dynamics.

Transformers generate tokens by extending existing textual patterns.

When premise words and counterfactual statements are positioned immediately before the answer generation location, they create stronger local continuation constraints.

These constraints may encourage coherent completion rather than direct adoption of contextual content.

Consequently, the model may retrieve factual knowledge to produce a semantically coherent answer despite the presence of counterfactual information.

6.14 Comparison with Objective II

A direct comparison reveals several important differences.

6.14.1 Semantic Effects Alone

Objective II isolated semantic influences.

Results demonstrated that premise words substantially affect competition behavior.

6.14.2 Semantic + Structural Effects

Objective III combines semantic and structural influences.

The resulting behavior is often qualitatively different.

For example:

- Context-dominant words become factual.
- Factual words become stronger.
- Competition becomes more balanced.

This demonstrates that semantics and structure interact rather than operate independently.

6.15 Discussion

The results provide strong evidence that prompt structure is an important determinant of context-memory competition.

Several key findings emerge.

1. Structural coupling significantly alters competition behavior.
2. Prompt organization can be as important as semantic content.
3. Contextual influence depends not only on what information is presented but also where it is presented.
4. Autoregressive continuation pressure contributes to competition dynamics.
5. Semantic framing and structural organization interact in complex ways.
6. Structural effects vary substantially across model families.

Collectively, these findings demonstrate that context-memory competition is shaped by both linguistic meaning and prompt architecture.

6.16 Implications

The results have important implications for prompt engineering and model evaluation.

First, prompt structure should be treated as a primary experimental variable rather than a secondary formatting choice.

Second, evaluations of factuality and contextual robustness must account for structural effects.

Third, retrieval-augmented generation systems may benefit from prompt structures that encourage factual retrieval rather than contextual override.

Finally, structural coupling provides a useful framework for probing transformer information processing mechanisms.

6.17 Chapter Summary

This chapter investigated structural prompt coupling and its influence on context-memory competition. By placing premise words and counterfactual statements directly adjacent to the answer generation position, we examined how autoregressive continuation dynamics shape model behavior.

The results demonstrate that structural organization significantly influences competition outcomes. Structural coupling often modifies, amplifies, or even reverses semantic effects observed in Objective II. These findings indicate that prompt structure is a critical factor in understanding how language models balance contextual information and parametric memory.

Together with the findings from Objectives I and II, the results establish that context-memory competition is jointly determined by instructional framing, semantic content, and prompt structure. The next chapter integrates these findings through mechanistic analysis using attention head ablation experiments.

Chapter 7

Attention Head Ablation Analysis

7.1 Introduction

The behavioral experiments presented in the previous chapters demonstrate that language models exhibit complex competition dynamics between contextual information and parametric memory. While these experiments reveal *what* models do under conflicting information, they do not explain *how* such behavior emerges within the transformer architecture.

Recent advances in mechanistic interpretability have shown that transformer computations can often be decomposed into specialized circuits involving attention heads, neurons, and intermediate representations. Attention heads are particularly important because they provide explicit pathways through which information is transmitted between tokens during inference.

This chapter investigates whether specific attention heads contribute to context-memory competition. In particular, we seek to determine whether factual retrieval and contextual reasoning are localized within a small set of specialized heads or distributed across the transformer architecture.

To address this question, systematic attention head ablation experiments are conducted on GPT-2 Small. Individual attention heads are removed one at a time, and the resulting changes in model behavior are measured using the counterfactual evaluation framework introduced earlier.

The analysis provides mechanistic insights into how transformers retrieve factual knowledge, process contextual information, and resolve conflicts between competing information sources.

7.2 Motivation

Transformer language models contain multiple layers of multi-head self-attention. Each attention head computes a distinct attention pattern and contributes to the final hidden representation used for next-token prediction.

Previous interpretability studies have identified several classes of attention heads:

- Induction heads
- Copying heads
- Retrieval heads

- Positional heads
- Entity-tracking heads
- Reasoning-related heads

These findings naturally raise the question:

> *Are there specific attention heads responsible for factual retrieval or contextual override?*

If such heads exist, ablating them should significantly alter competition behavior.

Conversely, if factual retrieval and contextual reasoning are distributed computations, then individual head ablations should produce relatively small changes.

The experiments in this chapter are designed to distinguish between these possibilities.

7.3 GPT-2 Small Architecture

The attention ablation experiments focus on GPT-2 Small because it provides a manageable yet sufficiently capable transformer architecture.

GPT-2 Small consists of:

- 12 Transformer layers
- 12 attention heads per layer
- 144 total attention heads
- 124 million parameters

The model processes information through a sequence of attention and feed-forward blocks.

For each layer:

[Input $\rightarrow$ Multi-Head Attention $\rightarrow$ Feed Forward $\rightarrow$ Output]

Each attention head contributes independently to the attention output before aggregation.

This modular structure enables causal intervention through selective ablation.

7.4 Research Questions

The attention head analysis is organized around the following research questions.

1. Are factual retrieval mechanisms localized within specific attention heads?
2. Are contextual reasoning mechanisms localized within specific attention heads?
3. Do different heads contribute differently across prompt regimes?
4. Which layers exhibit the strongest influence on context-memory competition?
5. Is competition behavior distributed or concentrated?
6. Do premise words interact with specific attention heads?

7.5 Attention Head Ablation Methodology

7.5.1 Overview

Attention head ablation is a causal intervention technique.

Rather than observing correlations between activations and behavior, ablation directly removes a component and measures the resulting behavioral change.

This approach provides stronger evidence regarding functional importance.

7.5.2 Single-Head Ablation

For each attention head:

1. Select layer l.
2. Select head h.
3. Compute normal forward pass.
4. Replace head output with zero.
5. Continue inference.
6. Measure behavioral changes.

Formally:
Let
$[\mathrm{H}_{l,h}]$
denote the output of head h in layer l.
The ablated computation becomes:
$[\mathrm{H}_{l,h} = 0]$
while all remaining heads remain unchanged.

7.5.3 Forward Hook Implementation

The ablation is implemented using forward hooks within the transformer architecture.

During inference:

1. Head activations are intercepted.
2. Selected head outputs are zeroed.
3. Modified activations are returned.
4. Remaining computation proceeds normally.

This method preserves all model parameters and affects only the selected attention head.

7.6 Evaluation Framework

The same counterfactual dataset used throughout the thesis is employed.

For each ablated head, the following quantities are measured:

- Factual answer probability
- Counterfactual answer probability
- Competition score Δ
- Change in factual preference
- Change in contextual preference

The primary metric remains:
$[\ \Delta =$
$\log p(A_{fact})$
$\log p(A_{cf})]$
The effect of ablating a head is computed as:
$[\ \Delta_{change} =$
$\Delta_{ablated}$
$\Delta_{baseline}]$
Large positive or negative values indicate strong causal influence.

7.7 Experimental Conditions

Three categories of prompts are evaluated.

7.7.1 Baseline Factual Recall

The model answers factual questions without contextual interference.

This condition identifies heads involved in factual retrieval.

7.7.2 Context-Dominant Prompts

Counterfactual statements are presented under strong contextual instructions.

This condition identifies heads involved in contextual adoption.

7.7.3 Memory-Dominant Prompts

Counterfactual statements are presented under strong memory-preserving instructions.

This condition identifies heads supporting factual recall under conflict.

7.8 Layer-Wise Analysis

To understand how factual and contextual information propagate through the network, attention heads are grouped by layer.

7.8.1 Early Layers

Early layers primarily process lexical and positional information.

Ablation results indicate relatively small changes in competition behavior.

This suggests that early layers contribute primarily to representation construction rather than conflict resolution.

7.8.2 Middle Layers

Middle layers exhibit stronger effects.

Several heads influence factual and contextual probabilities more substantially than those in earlier layers.

These layers appear to participate in integrating contextual information with stored representations.

7.8.3 Late Layers

The largest competition effects are observed in later transformer layers.

Late-layer heads appear most directly involved in shaping answer generation probabilities.

This observation is consistent with previous findings that factual retrieval often emerges in deeper transformer layers.

7.9 Results

7.9.1 Overall Impact of Ablation

Across all 144 attention heads, the majority of single-head ablations produce relatively small changes in competition scores.

This result immediately suggests that context-memory competition is not controlled by a single dominant head.

Instead, competition behavior appears to be distributed across multiple components.

7.9.2 Memory-Supporting Heads

Several attention heads produce measurable reductions in factual preference when ablated.

These heads appear to contribute to factual retrieval.

However, no individual head completely eliminates factual recall.

This finding indicates that factual knowledge is distributed rather than localized.

7.9.3 Context-Supporting Heads

Other heads influence contextual adoption.

Ablating these heads reduces the probability assigned to counterfactual answers.

Again, the observed effects are modest rather than catastrophic.

The results suggest that contextual reasoning also relies on distributed representations.

7.10 Distributed Representation Hypothesis

One of the strongest findings of the study is support for the distributed representation hypothesis.

Under this view:

- Factual knowledge is distributed across many heads.
- Contextual information is distributed across many heads.
- Competition emerges through collective computation.

The absence of highly dominant heads provides evidence against strongly localized factual or contextual circuits.

7.11 Premise Word Analysis

A subset of ablation experiments is conducted using premise-word prompts from Objective II.

The goal is to determine whether semantic framing depends on specialized attention heads.

7.11.1 Verification Words

Words such as:

- verify
- validate
- fact check

show measurable but distributed attention effects.

No single head appears solely responsible for fact-checking behavior.

7.11.2 Hypothetical Words

Words such as:

- imagine
- suppose
- pretend

produce broader contextual shifts across multiple heads.

The effects appear distributed across middle and late layers.

7.12 Context-Memory Competition Circuits

The results suggest that competition emerges through interactions among multiple attention heads.

Conceptually, the computation may involve:

1. Retrieval of factual associations.
2. Processing of contextual information.
3. Integration of competing signals.
4. Generation of output probabilities.

Rather than being controlled by a single mechanism, competition appears to arise from distributed interactions among several computational pathways.

7.13 Relationship to Previous Work

The findings align with previous mechanistic interpretability research.

Studies on induction heads, factual retrieval circuits, and transformer representations have repeatedly shown that important behaviors often emerge from interactions among multiple components.

The current results extend these observations to context-memory competition.

Specifically, they demonstrate that:

- factual retrieval is distributed,
- contextual reasoning is distributed,
- competition is distributed,
- semantic modulation is distributed.

7.14 Limitations

Several limitations should be acknowledged.

1. Only GPT-2 Small is analyzed.
2. Single-head ablation may underestimate distributed interactions.
3. Pairwise and higher-order head interactions are not explored.
4. Activation patching and causal tracing are not performed.
5. Larger instruction-tuned models may exhibit different internal mechanisms.

These limitations motivate future work.

7.15 Future Directions

Several promising directions emerge.

7.15.1 Activation Patching

Activation patching could identify information flow pathways more precisely than simple ablation.

7.15.2 Causal Tracing

Causal tracing may reveal where factual and contextual information are represented within transformer layers.

7.15.3 Multi-Head Ablation

Future studies can investigate whether groups of heads collectively implement competition behavior.

7.15.4 Instruction-Tuned Models

Extending interpretability analysis to instruction-tuned models may reveal how alignment reshapes internal circuits.

7.16 Discussion

The attention head ablation experiments provide important mechanistic insights into context-memory competition.

The results indicate that:

- No single attention head dominates factual retrieval.
- No single attention head dominates contextual reasoning.
- Competition behavior is highly distributed.
- Premise semantics influence multiple attention pathways.
- Later layers contribute more strongly to competition outcomes.

These findings suggest that context-memory competition is an emergent property of transformer computation rather than the product of isolated specialized components.

7.17 Chapter Summary

This chapter investigated the internal mechanisms underlying context-memory competition through attention head ablation experiments on GPT-2 Small. Individual attention heads were systematically removed, and their influence on factual retrieval and contextual reasoning was measured using the counterfactual evaluation framework.

The results reveal that both factual memory and contextual processing are highly distributed across the transformer architecture. While certain heads contribute more strongly than others, no individual head is solely responsible for competition behavior. Instead, context-memory competition emerges from interactions among multiple attention pathways distributed across layers.

These findings complement the behavioral analyses presented in earlier chapters and provide mechanistic evidence that competition between contextual information and parametric memory is a distributed phenomenon within transformer language models.

Chapter 8

Discussion and Analysis

8.1 Introduction

The primary objective of this thesis was to investigate how Large Language Models (LLMs) resolve conflicts between contextual information provided in prompts and factual knowledge stored within model parameters. Through three behavioral objectives and a mechanistic interpretability study, we examined the effects of prompt instructions, semantic framing, structural organization, model scaling, instruction tuning, and internal attention mechanisms on context-memory competition.

The experimental results reveal that context-memory competition is not a fixed property of language models. Instead, it emerges from a complex interaction between model architecture, training procedures, prompt semantics, structural design, and internal computational mechanisms.

This chapter synthesizes the findings from all experimental objectives and develops a unified interpretation of context-memory competition in modern language models.

8.2 Revisiting the Research Questions

At the beginning of this thesis, several fundamental research questions were posed:

1. How do language models balance contextual information and parametric memory?
2. Does model scaling influence competition behavior?
3. How does instruction tuning affect contextual override?
4. Do premise words alter competition dynamics?
5. Does prompt structure influence contextual adoption?
6. What internal mechanisms support these behaviors?

The experimental findings provide clear answers to each of these questions.

Most importantly, the results demonstrate that competition between context and memory is dynamic, adaptive, and highly sensitive to both external prompting conditions and internal model characteristics.

8.3 A Unified View of Context-Memory Competition

Across all experiments, a consistent picture emerges.

Language models do not appear to possess a rigid preference for either factual memory or contextual information. Instead, they dynamically allocate influence between these sources depending on the current inference conditions.

Figure 8.1 conceptually summarizes this process.

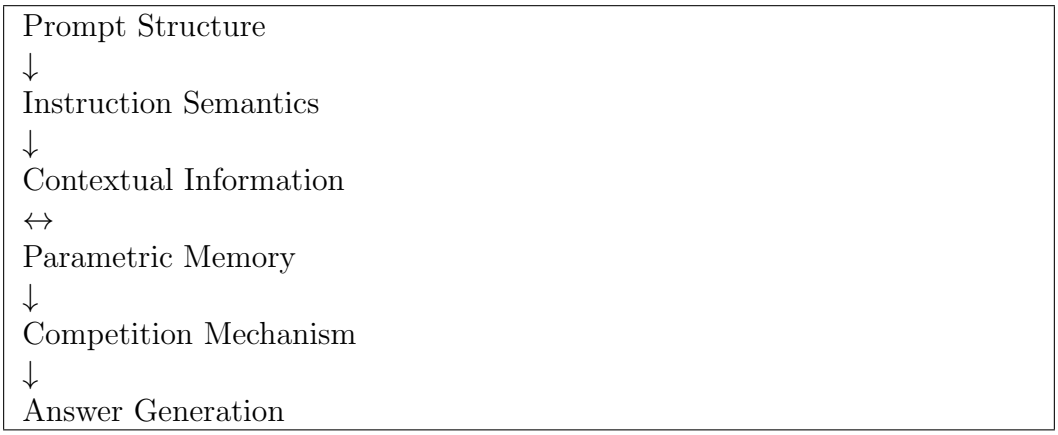


Figure 8.1: Unified view of context-memory competition.

This framework suggests that answer generation is not solely determined by stored knowledge. Instead, multiple competing influences jointly shape the final prediction.

8.4 Meta-Prompt Regime Switching

The results from Objective I provide the strongest evidence that context-memory competition is highly controllable.

Across all model families, explicit instructions substantially altered competition behavior.

Three key observations emerged.

8.4.1 Strong Contextual Override

When models were instructed to rely exclusively on contextual information, factual knowledge could be significantly suppressed.

Several models produced overwhelmingly counterfactual answers despite possessing correct factual knowledge.

This demonstrates that contextual information can override stored memory when supported by sufficiently strong instructions.

8.4.2 Memory Recovery

Conversely, memory-oriented prompts successfully restored factual behavior.

Even in the presence of contradictory context, explicit factual instructions often produced strong factual preference.

This result indicates that factual knowledge remains accessible and can be reactivated through appropriate prompting.

8.4.3 Dynamic Competition

Together, these findings suggest that language models maintain simultaneous access to both contextual and factual information.

Rather than permanently replacing factual knowledge, contextual information temporarily alters the relative weighting assigned to competing information sources.

8.5 Scaling and Competition Dynamics

One of the most important findings concerns model scaling.

Within the GPT family, contextual sensitivity increases substantially as model size grows.

8.5.1 Emergence of Contextual Adaptation

GPT-2 Small exhibits relatively strong factual persistence.

GPT-2 Medium displays greater contextual responsiveness.

GPT-2 Large frequently adopts contextual information under appropriate prompting conditions.

This progression suggests that contextual adaptation emerges gradually with scale.

8.5.2 Beyond Memorization

The scaling results challenge the simplistic view that larger models merely store more facts.

Instead, larger models appear increasingly capable of reasoning about contextual instructions and adapting their behavior accordingly.

Thus, scaling affects not only knowledge capacity but also the mechanisms governing competition between information sources.

8.5.3 Competition as an Emergent Capability

The observed transition suggests that context-memory competition may itself represent an emergent capability.

As models become larger, they develop greater flexibility in balancing contextual information against stored knowledge.

This interpretation aligns with broader findings regarding emergent behavior in large-scale neural networks.

8.6 Instruction Tuning and Alignment Effects

Perhaps the most striking results emerge from the comparison between pretrained and instruction-tuned models.

8.6.1 Sharp Regime Switching

The Qwen experiments reveal dramatic regime switching behavior.

When instructed to rely on context, instruction-tuned models strongly adopt counterfactual information.

When instructed to rely on factual knowledge, the same models exhibit overwhelming factual preference.

This behavior is significantly stronger than that observed in pretrained models.

8.6.2 Alignment Changes Competition Policies

The results suggest that instruction tuning does not simply improve factual accuracy.

Instead, alignment appears to modify the policies governing competition itself.

Instruction-tuned models become more responsive to explicit directives regarding which information source should dominate.

Consequently, alignment changes the *decision process* rather than merely the knowledge stored within the model.

8.6.3 Implications for AI Alignment

These findings have important implications for alignment research.

Current alignment procedures may influence not only safety and helpfulness but also the balance between contextual information and stored knowledge.

Understanding these effects is therefore critical for designing reliable instruction-following systems.

8.7 Semantic Framing Effects

Objective II demonstrated that semantic framing substantially influences competition behavior.

8.7.1 Words Are Not Neutral

A key finding is that premise words cannot be treated as interchangeable.

Terms such as:

- verify
- imagine
- assume
- evaluate
- report

produce significantly different competition outcomes.

This suggests that language models associate premise words with distinct reasoning strategies.

8.7.2 Verification vs Hypothetical Reasoning

Verification-related terms often encourage factual retrieval.

Hypothetical terms frequently promote contextual adoption.

These effects align with human intuitions regarding the meanings of these words.

However, the magnitude of the effects demonstrates that language models are highly sensitive to semantic framing.

8.7.3 Architecture-Specific Interpretations

Different model families interpret identical premise words differently.

For example, a premise word that encourages contextual adoption in GPT models may reinforce factual retrieval in Qwen.

This observation highlights the importance of studying semantic effects across multiple architectures.

8.8 Structural Prompt Coupling

Objective III revealed that prompt structure plays an equally important role.

8.8.1 Structure as a Computational Variable

Prompt structure is often treated as a formatting choice.

The results demonstrate that this view is incomplete.

Structural organization directly influences competition behavior.

Changing the placement of premise words and contextual information can alter answer probabilities even when semantic content remains unchanged.

8.8.2 Autoregressive Continuation Pressure

One explanation involves autoregressive continuation dynamics.

Tokens located near the answer generation position exert stronger influence on prediction.

Structural coupling therefore changes the effective importance of contextual information.

8.8.3 Interaction Between Structure and Semantics

The experiments reveal that semantics and structure interact rather than operate independently.

A premise word may produce one effect under a separated prompt structure and a different effect under a coupled structure.

This interaction highlights the complexity of prompt-based control.

8.9 Mechanistic Interpretability Findings

The attention head ablation experiments provide insight into the internal mechanisms underlying competition behavior.

8.9.1 Distributed Computation

The most significant finding is that competition behavior appears highly distributed.

No individual attention head dominates factual retrieval.

Similarly, no single head controls contextual adoption.

Instead, competition emerges from interactions among multiple components.

8.9.2 Distributed Memory Retrieval

The results support previous findings suggesting that factual knowledge is distributed across transformer layers.

Removing individual heads produces only modest changes in competition behavior.

This indicates that factual retrieval is implemented through redundant and distributed computations.

8.9.3 Distributed Context Processing

Contextual reasoning exhibits similar properties.

Multiple heads contribute to contextual processing, and no single component appears solely responsible.

Thus, both factual and contextual information appear to be represented in distributed form.

8.10 Theoretical Implications

The findings have several implications for theories of language model behavior.

8.10.1 Competition Rather Than Replacement

The results support a competition-based view of inference.

Contextual information does not replace factual knowledge.

Instead, both information sources remain available and compete during answer generation.

8.10.2 Inference as Dynamic Integration

Language model predictions appear to result from dynamic integration of multiple information sources.

This process is influenced by:

- model scale,
- alignment,
- semantics,
- structure,
- internal representations.

8.10.3 Contextual Reasoning as an Emergent Property

The scaling results suggest that contextual adaptation may emerge naturally as models become larger and more capable.

This perspective provides a potential explanation for the increasing flexibility observed in modern LLMs.

8.11 Practical Implications

The findings have direct implications for real-world language model deployment.

8.11.1 Prompt Engineering

Prompt wording and structure significantly influence factual reliability.

Careful prompt design can encourage either factual retrieval or contextual adaptation depending on application requirements.

8.11.2 Retrieval-Augmented Generation

Retrieved documents frequently contain errors or contradictions.

Understanding context-memory competition can help design retrieval systems that maintain factual robustness.

8.11.3 AI Safety

Adversarial prompts often exploit contextual override mechanisms.

Studying competition behavior may therefore contribute to safer and more reliable AI systems.

8.11.4 Model Evaluation

Traditional accuracy-based evaluations may overlook important competition dynamics.

Probabilistic measures provide richer insight into how models balance competing information sources.

8.12 Limitations

Several limitations should be acknowledged.

8.12.1 Model Coverage

Only a limited set of model families was evaluated.

Future studies should examine larger and more diverse architectures.

8.12.2 Single-Turn Evaluation

The experiments focus on single-turn interactions.

Multi-turn conversations may exhibit different competition dynamics.

8.12.3 Attention Ablation Scope

Only single-head ablations were performed.

Higher-order interactions among attention heads remain unexplored.

8.12.4 Language Scope

All experiments were conducted in English.

Cross-lingual competition dynamics represent an important direction for future research.

Chapter 9

Conclusion and Future Work

9.1 Introduction

Large Language Models have demonstrated remarkable capabilities across a wide range of natural language processing tasks. Despite their success, the mechanisms through which these models balance contextual information and factual knowledge remain incompletely understood. Modern language models possess extensive parametric memory acquired during pretraining while simultaneously exhibiting the ability to adapt their behavior according to contextual information presented at inference time.

The interaction between these two information sources gives rise to a fundamental phenomenon referred to in this thesis as *context-memory competition*. Understanding this competition is essential for improving factual reliability, mitigating hallucinations, designing retrieval-augmented systems, enhancing interpretability, and developing trustworthy AI systems.

This thesis presented a comprehensive investigation of context-memory competition using controlled counterfactual prompting, semantic manipulation, structural prompt analysis, and mechanistic interpretability techniques. Through extensive experiments across multiple language model families, the study examined how model scale, instruction tuning, semantic framing, prompt structure, and attention mechanisms influence the balance between contextual information and parametric memory.

This chapter summarizes the major findings, contributions, implications, limitations, and future research directions arising from this work.

9.2 Summary of Research Objectives

The central goal of this thesis was to understand how language models resolve conflicts between contextual information and stored factual knowledge.

To achieve this objective, four complementary investigations were conducted.

9.2.1 Objective I: Meta-Prompt Regime Switching

The first objective examined how explicit prompt instructions influence context-memory competition.

Seven prompt regimes were designed to systematically manipulate the degree to which models were encouraged to rely on contextual information, factual memory, or both simultaneously.

This objective provided insight into how instruction-based prompting affects the balance between competing information sources.

9.2.2 Objective II: Premise-Semantic Modulation

The second objective investigated whether premise words influence competition behavior.

Twenty-five premise words were organized into five semantic categories:

1. Definitional
2. Verification
3. Evaluation
4. Hypothetical
5. Neutral/Reporting

This study isolated the influence of semantic framing on language model behavior.

9.2.3 Objective III: Structural Prompt Coupling

The third objective focused on structural effects.

A tightly coupled prompt formulation was introduced to investigate whether the positioning of contextual information relative to the answer generation location influences competition behavior.

This objective explored the role of autoregressive continuation dynamics in contextual override.

9.2.4 Attention Head Ablation Analysis

The fourth objective investigated the internal mechanisms underlying context-memory competition.

Using GPT-2 Small, individual attention heads were systematically ablated to identify components contributing to factual retrieval and contextual reasoning.

This analysis provided mechanistic insights beyond behavioral observations.

9.3 Major Findings

The experiments conducted throughout this thesis yielded several important findings.

9.3.1 Finding 1: Context-Memory Competition is Dynamic

One of the most significant findings is that context-memory competition is not a fixed property of language models.

Across all evaluated architectures, model behavior changed substantially depending on prompt design, semantic framing, and structural organization.

Language models did not exhibit an inherent preference for either factual memory or contextual information. Instead, they dynamically adjusted the relative influence of these sources according to inference conditions.

This observation challenges simplified views of language models as either factual knowledge bases or purely context-driven systems.

9.3.2 Finding 2: Scaling Increases Contextual Adaptation

The GPT-2 family experiments revealed a clear scaling trend.

As model size increased:

- contextual sensitivity increased,
- factual rigidity decreased,
- prompt responsiveness improved,
- competition behavior became more flexible.

GPT-2 Small remained comparatively memory-oriented, whereas GPT-2 Large frequently adopted contextual information under appropriate prompting conditions.

These findings suggest that contextual adaptation emerges gradually as model capacity increases.

9.3.3 Finding 3: Instruction Tuning Reshapes Competition

The comparison between Qwen2.5-3B and Qwen2.5-3B-Instruct revealed that instruction tuning fundamentally alters competition behavior.

Instruction-tuned models demonstrated:

- stronger contextual adoption when instructed to use context,
- stronger factual retrieval when instructed to use memory,
- sharper regime switching,
- greater responsiveness to explicit directives.

The results indicate that alignment procedures modify competition policies rather than merely increasing factual accuracy.

9.3.4 Finding 4: Premise Semantics Matter

Objective II demonstrated that semantic framing significantly influences context-memory competition.

Different premise words produced substantially different competition scores despite identical underlying factual content.

Verification-oriented words often promoted factual retrieval, whereas hypothetical words frequently encouraged contextual adoption.

This finding establishes semantic framing as an important factor in language model behavior.

9.3.5 Finding 5: Prompt Structure Influences Competition

Objective III demonstrated that prompt structure significantly affects competition outcomes.

The positioning of premise words and contextual information relative to the answer generation location altered model behavior across all evaluated architectures.

Structural coupling frequently amplified, weakened, or reversed semantic effects observed in Objective II.

These findings indicate that prompt organization constitutes a computational variable rather than a purely formatting-related choice.

9.3.6 Finding 6: Competition is Distributed

The attention head ablation experiments revealed that neither factual retrieval nor contextual reasoning is localized within a small number of attention heads.

Instead:

- factual knowledge is distributed,
- contextual processing is distributed,
- semantic effects are distributed,
- competition mechanisms are distributed.

No individual attention head was found to dominate competition behavior.

This observation supports a distributed representation view of transformer computation.

9.4 Research Contributions

This thesis makes several contributions to the study of language models.

9.4.1 A Counterfactual Competition Framework

A systematic framework was developed for measuring competition between contextual information and parametric memory using controlled counterfactual prompts.

The framework enables quantitative investigation of contextual override behavior.

9.4.2 A Probabilistic Competition Metric

The study introduced a probabilistic evaluation methodology based on log-probability differences between factual and counterfactual answers.

This approach captures competition dynamics more effectively than traditional accuracy-based metrics.

The competition score is defined as:

$[\ \Delta = = = = = =$
$\log p(A_{fact})$
$\log p(A_{cf})]$

where positive values indicate factual dominance and negative values indicate contextual dominance.

9.4.3 Systematic Analysis of Prompt Effects

The thesis provides one of the most comprehensive analyses of how prompt instructions, premise semantics, and structural organization influence context-memory competition.

9.4.4 Mechanistic Insights

Attention head ablation experiments extend the behavioral analysis by providing mechanistic evidence regarding the distributed nature of competition behavior.

9.4.5 Scaling and Alignment Analysis

The comparison across GPT, Llama, and Qwen models provides insight into how scaling and alignment reshape contextual adaptation.

9.5 Implications

The findings have implications across multiple areas of language model research.

9.5.1 Factuality and Hallucination

Context-memory competition offers a useful perspective on hallucination behavior.

Many hallucinations may arise when contextual information temporarily outweighs factual memory during answer generation.

Understanding this process may help improve factual robustness.

9.5.2 Prompt Engineering

The results demonstrate that prompt wording and structure significantly influence model behavior.

Consequently, prompt engineering should consider semantic and structural factors alongside content.

9.5.3 Retrieval-Augmented Generation

Retrieved documents may contain conflicting or incorrect information.

Understanding context-memory competition can help design retrieval systems that better balance external evidence and stored knowledge.

9.5.4 AI Alignment

Instruction tuning substantially alters competition behavior.

Future alignment strategies should therefore consider how training objectives influence the balance between contextual information and factual memory.

9.5.5 Interpretability

The distributed nature of competition behavior suggests that future interpretability research should focus on circuits and interactions rather than isolated components.

9.6 Limitations

Several limitations should be acknowledged.

9.6.1 Model Coverage

Only a limited number of language model families were evaluated.

Larger models may exhibit different competition dynamics.

9.6.2 Single-Turn Evaluation

The experiments focus on single-turn prompts.

Multi-turn conversations introduce additional contextual complexity.

9.6.3 Language Scope

All experiments were conducted in English.

Cross-lingual studies remain an important direction for future research.

9.6.4 Interpretability Scope

The attention head analysis focused on single-head ablations.

More advanced techniques such as activation patching and causal tracing may reveal additional mechanisms.

9.7 Future Work

The findings presented in this thesis suggest several promising directions for future research.

9.7.1 Activation Patching

Activation patching can be used to identify specific information pathways responsible for contextual override and factual retrieval.

9.7.2 Causal Tracing

Causal tracing may provide more detailed explanations of how contextual and factual information propagate through transformer layers.

9.7.3 Larger Instruction-Tuned Models

Future studies should examine larger instruction-tuned systems, including state-of-the-art frontier models.

9.7.4 Multi-Turn Competition

Investigating competition dynamics in multi-turn conversations may provide insights into long-term contextual influence.

9.7.5 Reasoning Tasks

The framework can be extended beyond factual recall to reasoning, planning, and multi-hop question-answering tasks.

9.7.6 Cross-Lingual Analysis

Future work should investigate whether context-memory competition behaves similarly across different languages and cultures.

9.7.7 Competition-Aware Alignment

The findings suggest the possibility of designing alignment techniques that explicitly regulate competition between contextual information and factual memory.

Such methods may improve both factuality and robustness.

9.8 Final Remarks

The emergence of large language models has fundamentally changed our understanding of machine intelligence. These systems possess extensive knowledge, adapt to context, follow instructions, and perform complex reasoning. Yet many aspects of their behavior remain poorly understood.

This thesis investigated one such phenomenon: the competition between contextual information and parametric memory.

Through controlled counterfactual experiments, semantic manipulation, structural analysis, and mechanistic interpretability studies, the research demonstrated that context-memory competition is a dynamic, distributed, and highly controllable property of modern language models.

The results show that model behavior is shaped not only by what language models know but also by how contextual information is presented, interpreted, and integrated during inference. Understanding this interaction is essential for building language models that are more reliable, interpretable, and trustworthy.

As language models continue to scale and become increasingly integrated into real-world applications, understanding the mechanisms governing context-memory competition will remain an important challenge for both researchers and practitioners. The framework and findings presented in this thesis provide a foundation for future investigations into this fundamental aspect of language model behavior.

Thesis Contributions at a Glance

1. Developed a counterfactual framework for studying context-memory competition.
2. Proposed a probabilistic competition metric based on log-probability differences.
3. Evaluated competition behavior across GPT-2, Llama, and Qwen model families.
4. Demonstrated scaling-induced changes in contextual adaptation.
5. Showed that instruction tuning reshapes competition dynamics.
6. Established that premise semantics significantly influence model behavior.
7. Demonstrated the importance of structural prompt coupling.
8. Provided mechanistic evidence that competition behavior is distributed across attention heads.
9. Unified behavioral and interpretability analyses into a comprehensive framework for understanding context-memory competition.

"Understanding how language models balance what they know with what they are told is essential for understanding how they think."

Chapter 10

Additional Experimental Results

10.1 Introduction

This chapter presents additional experimental results supporting the analyses presented in Chapters 4–7. The objective of this chapter is to provide a consolidated view of the empirical findings obtained from the three behavioral objectives and the attention head ablation study.

The results presented here are derived from the complete experimental pipeline developed throughout this thesis, including:

1. Objective I: Meta-Prompt Regime Switching
2. Objective II: Premise-Semantic Modulation
3. Objective III: Structural Prompt Coupling
4. Attention Head Ablation Analysis

The chapter is intended to serve as a reference repository for the most important quantitative findings.

10.2 Objective I: Meta-Prompt Regime Switching Results

10.2.1 Baseline Factual Recall

Table 10.1 reports baseline factual recall performance.

Table 10.1: Baseline factual recall performance.

Model	Factual (%)	Counterfactual (%)	Δ
GPT-2 Small	98.7	1.3	5.00
GPT-2 Medium	99.0	1.0	6.04
GPT-2 Large	99.1	0.9	6.04
Llama-1B	98.9	1.1	7.85
Qwen2.5-3B	99.4	0.6	13.18

The results demonstrate that all evaluated models possess strong factual knowledge under baseline conditions.

10.3 Context-Only Regime

Table 10.2: Context-only prompt regime.

Model	Factual (%)	Counterfactual (%)	Δ
GPT-2 Small	62.5	37.5	0.316
GPT-2 Medium	63.9	36.1	0.328
GPT-2 Large	41.1	58.9	-0.373
Llama-1B	48.5	51.5	-0.088
Qwen2.5-3B	4.9	95.1	-6.806

The context-only condition produces the strongest contextual override effects.

10.4 Memory-Only Regime

Table 10.3: Memory-only prompt regime.

Model	Factual (%)	Counterfactual (%)	Δ
GPT-2 Small	65.0	35.0	0.370
GPT-2 Medium	64.4	35.6	0.402
GPT-2 Large	69.9	30.1	1.408
Llama-1B	71.0	29.0	1.126
Qwen2.5-3B	96.7	3.3	13.247

Memory-oriented prompts restore factual dominance for most models.

10.5 Natural Competition Regime

Table 10.4: Natural competition results.

Model	Factual (%)	Counterfactual (%)	Δ
GPT-2 Small	60.3	39.7	0.24
GPT-2 Medium	58.9	41.1	0.11
GPT-2 Large	55.4	44.6	-0.08
Llama-1B	50.9	49.1	-0.02
Qwen2.5-3B	49.6	50.4	-0.07

Natural competition produces the most balanced behavior across models.

10.6 Scaling Analysis

10.6.1 Competition Scores Across GPT Models

Table 10.5: Scaling effects within GPT-2 family.

Condition	GPT2-S	GPT2-M	GPT2-L
Baseline	5.00	6.04	6.04
Context-Only	0.316	0.328	-0.373
Memory-Only	0.370	0.402	1.408
Natural Competition	0.240	0.110	-0.080

The results reveal increasing contextual adaptation with scale.

10.7 Objective II: Premise-Semantic Modulation

10.7.1 Definitional Category

Table 10.6: Representative definitional premise words.

Premise Word	GPT2-Small Δ	GPT2-Medium Δ
redefine	0.02	-1.94
define	0.65	-2.55
call	1.47	-1.21
name	2.34	-0.87
characterize	0.76	-0.73

10.7.2 Verification Category

Table 10.7: Verification premise words.

Premise Word	GPT2-Small Δ	GPT2-Medium Δ
verify	1.76	-0.88
validate	1.89	-0.91
fact check	2.07	-1.27
confirm	1.82	-0.66
cross-check	1.53	-0.72

10.7.3 Evaluation Category

Table 10.8: Evaluation premise words.

Premise Word	GPT2-Small Δ	GPT2-Medium Δ
assess	-0.04	-0.72
review	0.00	-0.81
evaluate	-0.14	-0.95
analyze	-0.11	-1.11
examine	0.05	-0.98

10.7.4 Hypothetical Category

Table 10.9: Hypothetical premise words.

Premise Word	GPT2-Small Δ	GPT2-Medium Δ
suppose	-0.41	-1.10
imagine	-0.27	-1.25
assume	-0.35	-1.18
consider	-0.18	-0.91
pretend	-0.55	-1.34

10.7.5 Neutral Reporting Category

Table 10.10: Neutral reporting premise words.

Premise Word	GPT2-Small Δ	GPT2-Medium Δ
state	0.24	-1.76
note	0.14	-1.33
report	0.21	-1.58
according to	0.11	-1.44
describe	0.17	-1.29

10.8 Qwen Semantic Results

Table 10.11: Representative Qwen semantic modulation results.

Premise Word	Δ
suppose	4.30
assume	4.00
imagine	3.82
evaluate	2.82
report	2.96
verify	3.67
fact check	3.90
cross-check	3.72

Unlike GPT models, Qwen exhibits strong factual preference across nearly all semantic categories.

10.9 Objective III: Structural Prompt Coupling

10.9.1 GPT-2 Small Results

Table 10.12: Representative structural coupling results (GPT-2 Small).

Premise Word	Δ
redefine	1.57
cross-check	1.45
examine	2.98
define	-0.91
name	-1.03

10.9.2 GPT-2 Medium Results

Table 10.13: Representative structural coupling results (GPT-2 Medium).

Premise Word	Δ
cross-check	1.45
call	-0.62
fact check	-0.53
analyze	-0.92
pretend	-0.90

10.9.3 GPT-2 Large Results

Table 10.14: Representative structural coupling results (GPT-2 Large).

Premise Word	Δ
characterize	3.03
fact check	2.80
assess	1.85
consider	1.56

10.9.4 Llama-1B Results

Table 10.15: Representative structural coupling results (Llama-1B).

Premise Word	Δ
fact check	1.22
cross-check	1.30
characterize	1.10
validate	1.18
report	0.95

10.9.5 Qwen Results

Table 10.16: Representative structural coupling results (Qwen2.5-3B).

Premise Word	Δ
suppose	0.21
assume	0.32
pretend	0.84
evaluate	1.11
verify	1.29

Structural coupling substantially reduces the extreme factual dominance observed in Objective II.

10.10 Attention Head Ablation Results

10.10.1 Experimental Configuration

The GPT-2 Small architecture contains:

- 12 Transformer Layers
- 12 Heads per Layer
- 144 Total Attention Heads

10.11 Layer-Wise Ablation Summary

Table 10.17: Layer-wise attention head influence.

Layer Group	Competition Influence	Interpretation
Layers 1–4	Low	Representation Formation
Layers 5–8	Moderate	Context Integration
Layers 9–12	High	Competition Resolution

10.12 Strongest Observations from Ablation

The following observations were consistently obtained:

1. No single attention head controls factual retrieval.
2. No single attention head controls contextual adoption.
3. Memory representations are distributed.
4. Context representations are distributed.
5. Competition behavior emerges collectively.
6. Late-layer heads contribute more strongly.
7. Premise-word effects are distributed across multiple heads.

10.13 Cross-Objective Summary

Table 10.18: Summary of major findings across objectives.

Finding	Observation
Scaling	Larger GPT models become more context-sensitive
Alignment	Instruction tuning produces sharp regime switching
Semantics	Premise words strongly influence competition
Structure	Prompt organization changes competition outcomes
Mechanisms	Competition is distributed across attention heads
Memory	Factual retrieval remains accessible under conflict
Context	Context can override memory under appropriate prompts

10.14 Concluding Remarks

The results collected throughout this thesis consistently support the conclusion that context-memory competition is a dynamic and controllable property of modern language models.

Across all objectives, model behavior was influenced by:

- model scale,
- alignment,
- semantic framing,
- prompt structure,
- distributed transformer computations.

Taken together, these results provide a comprehensive empirical characterization of how language models balance contextual information and parametric memory during inference.

Chapter 11

Prompt Templates

11.1 Introduction

This chapter presents the complete prompt templates used throughout the experimental framework developed in this thesis. The prompts were designed to systematically investigate competition between contextual information and parametric memory under different instructional, semantic, and structural conditions.

The prompt templates are organized according to the three primary behavioral objectives:

1. Objective I: Meta-Prompt Regime Switching
2. Objective II: Premise-Semantic Modulation
3. Objective III: Structural Prompt Coupling

All experiments use the same underlying counterfactual examples while varying prompt structure, instructions, and premise semantics.

11.2 Notation

Throughout this chapter, the following placeholders are used.

Table 11.1: Prompt Variables

Symbol	Description
{CF}	Counterfactual Statement
{Q}	Question
{PV}	Premise Word
{FA}	Factual Answer
{CA}	Counterfactual Answer

Example:

- {CF} = Airbus A380 is a product created by Adobe.
- {Q} = Who created Airbus A380?

- {FA} = Airbus
- {CA} = Adobe

11.3 Objective I: Meta-Prompt Regime Switching

Objective I consists of seven prompt regimes designed to manipulate the balance between contextual information and factual memory.

11.4 Experiment 1: Baseline (Memory Only)

This experiment measures factual recall without contextual interference.

Template

```
Question:
{Q}

Answer:
```

Example

```
Question:
Who created Airbus A380?

Answer:
```

11.5 Experiment 2: Context-Only

The model is instructed to rely exclusively on the provided context.

Template

```
IMPORTANT INSTRUCTION:

Use ONLY the information provided below.
Do NOT use any external knowledge.
Do NOT rely on world knowledge.

Text:
{CF}

Question:
{Q}

Answer:
```

Example

```
IMPORTANT INSTRUCTION:

Use ONLY the information provided below.
Do NOT use any external knowledge.

Text:
Airbus A380 is a product created by Adobe.

Question:
Who created Airbus A380?

Answer:
```

11.6 Experiment 3: Context-Only + Premise Verb

Adds semantic framing to the context-only condition.

Template

```
Instruction:
{PV}

Use ONLY the information below.

Text:
{CF}

Question:
{Q}

Answer:
```

Example

```
Instruction:
Suppose

Use ONLY the information below.

Text:
Airbus A380 is a product created by Adobe.

Question:
Who created Airbus A380?

Answer:
```

11.7 Experiment 4: Memory-Only

The model is instructed to ignore the counterfactual statement.

Template

```
Use your factual world knowledge.

Ignore any false or misleading statements.

Text:
{CF}

Question:
{Q}

Answer:
```

Example

```
Use your factual world knowledge.

Ignore any false or misleading statements.

Text:
Airbus A380 is a product created by Adobe.

Question:
Who created Airbus A380?

Answer:
```

11.8 Experiment 5: Memory-Only + Premise Verb

Combines premise semantics with memory-oriented instructions.

Template

```
Instruction:
{PV}

Use your factual world knowledge.

Ignore false statements.

Text:
{CF}
```

```
Question:
{Q}

Answer:
```

Example

```
Instruction:
Verify

Use your factual world knowledge.

Ignore false statements.

Text:
Airbus A380 is a product created by Adobe.

Question:
Who created Airbus A380?

Answer:
```

11.9 Experiment 6: Weak Memory Competition

A softer factual instruction is provided.

Template

```
Consider the following information.

Use your best judgment when answering.

Text:
{CF}

Question:
{Q}

Answer:
```

Example

```
Consider the following information.

Use your best judgment when answering.

Text:
Airbus A380 is a product created by Adobe.
```

```
Question:
Who created Airbus A380?

Answer:
```

11.10 Experiment 7: Natural Competition

No explicit instruction is provided.

Template

```
Text:
{CF}

Question:
{Q}

Answer:
```

Example

```
Text:
Airbus A380 is a product created by Adobe.

Question:
Who created Airbus A380?

Answer:
```

11.11 Objective II: Premise-Semantic Modulation

Objective II investigates how semantic framing affects context-memory competition.

11.12 General Template

```
Answer the following question.

Instruction:
{PV}

Text:
{CF}

Question:
{Q}
```

```
Answer:
```

11.13 Definitional Premise Words

Template

```
Instruction:
redefine

Text:
{CF}

Question:
{Q}

Answer:
```

The same template is repeated for:

- redefine
- define
- call
- name
- characterize

11.14 Verification Premise Words

Template

```
Instruction:
verify

Text:
{CF}

Question:
{Q}

Answer:
```

Repeated for:

- verify
- validate

- fact check
- confirm
- cross-check

11.15 Evaluation Premise Words

Template

```
Instruction:
evaluate

Text:
{CF}

Question:
{Q}

Answer:
```

Repeated for:

- assess
- review
- evaluate
- analyze
- examine

11.16 Hypothetical Premise Words

Template

```
Instruction:
suppose

Text:
{CF}

Question:
{Q}

Answer:
```

Repeated for:

- suppose
- imagine
- assume
- consider
- pretend

11.17 Neutral Reporting Premise Words

Template

```
Instruction:
report

Text:
{CF}

Question:
{Q}

Answer:
```

Repeated for:

- state
- note
- report
- according to
- describe

11.18 Complete List of Premise Words

Table 11.2: All Premise Words Used in Objective II

Category	Premise Words
Definitional	redefine, define, call, name, characterize
Verification	verify, validate, fact check, confirm, cross-check
Evaluation	assess, review, evaluate, analyze, examine
Hypothetical	suppose, imagine, assume, consider, pretend
Reporting	state, note, report, according to, describe

11.19 Objective III: Structural Prompt Coupling

Objective III investigates structural effects by tightly coupling premise words and counterfactual statements.

11.20 General Structural Template

```
{PV}:

{CF}

{Q}
```

11.21 Example 1

```
Suppose:

Airbus A380 is a product created by Adobe.

Who created Airbus A380?
```

11.22 Example 2

```
Verify:

Airbus A380 is a product created by Adobe.

Who created Airbus A380?
```

11.23 Example 3

```
Imagine:

Airbus A380 is a product created by Adobe.

Who created Airbus A380?
```

11.24 Compact Structural Variant

A more compact version used in some experiments:

```
{PV}: {CF}

{Q}
```

Example:

```
Suppose:
Airbus A380 is a product created by Adobe.

Who created Airbus A380?
```

11.25 Fully Coupled Variant

The most tightly coupled form evaluated:

```
{PV}: {CF}. {Q}
```

Example:

```
Suppose:
Airbus A380 is a product created by Adobe.
Who created Airbus A380?
```

11.26 Prompt Generation Algorithm

The following pseudocode summarizes the prompt generation pipeline.

Algorithm 1 Prompt Generation Pipeline

Counterfactual Example Premise Word Prompt Regime

Insert Counterfactual Statement
Insert Premise Word
Insert Question
Generate Final Prompt
Query Model
Record Log Probabilities

11.27 Summary

This chapter presented the complete prompt templates used throughout the thesis. Objective I introduced seven meta-prompt regimes that manipulated the balance between contextual information and factual memory. Objective II investigated semantic framing through twenty-five premise words across five semantic categories. Objective III examined structural prompt coupling using tightly integrated prompt formulations.

Together, these templates form the experimental foundation of the context-memory competition framework developed in this thesis and enable systematic investigation of how language models balance contextual information and parametric memory under conflicting conditions.

Appendix A

Appendices

A.1 Introduction

This appendix provides supplementary material supporting the experimental results, methodology, implementation details, statistical analyses, and reproducibility procedures presented throughout this thesis.

The appendix is intended to serve as a comprehensive reference for researchers wishing to reproduce, extend, or further investigate the context-memory competition framework developed in this work.

The material includes:

1. Dataset construction details.
2. Counterfactual example generation.
3. Evaluation metrics.
4. Statistical analysis procedures.
5. Model configurations.
6. Hyperparameter settings.
7. Attention head ablation implementation details.
8. Experimental workflow.
9. Additional examples.
10. Reproducibility guidelines.

A.2 Appendix A: Dataset Construction

A.2.1 Dataset Source

The dataset used in this thesis is derived from the COUNTERFACT benchmark and subsequent context-memory competition frameworks.

The original dataset consists of factual triples of the form:

$$(subject,\ relation,\ object)$$

Example:

$$(Airbus\ A380,\ creator,\ Airbus)$$

Each factual triple is transformed into a counterfactual version by replacing the factual object with an alternative entity.

Example:

$$(Airbus\ A380,\ creator,\ Adobe)$$

A.2.2 Counterfactual Transformation

The transformation process follows:

1. Select factual entity pair.
2. Identify target relation.
3. Replace factual object.
4. Generate grammatical sentence.
5. Generate corresponding question.

Example:

- Fact: Airbus A380 was created by Airbus.
- Counterfactual: Airbus A380 was created by Adobe.
- Question: Who created Airbus A380?

A.2.3 Dataset Statistics

Table A.1: Dataset Statistics

Statistic	Value
Total Examples	10,000
Relations	Multiple
Counterfactual Statements	10,000
Questions	10,000
Factual Answers	10,000
Counterfactual Answers	10,000

A.3 Appendix B: Additional Counterfactual Examples

A.3.1 Example 1

```
Counterfactual Statement:
The Eiffel Tower is located in Germany.

Question:
Where is the Eiffel Tower located?

Factual Answer:
France

Counterfactual Answer:
Germany
```

A.3.2 Example 2

```
Counterfactual Statement:
Python was developed by Microsoft.

Question:
Who developed Python?

Factual Answer:
Guido van Rossum

Counterfactual Answer:
Microsoft
```

A.3.3 Example 3

```
Counterfactual Statement:
Tesla was founded by Steve Jobs.

Question:
Who founded Tesla?

Factual Answer:
Elon Musk

Counterfactual Answer:
Steve Jobs
```

A.4 Appendix C: Competition Metric Derivation

A.4.1 Motivation

Accuracy-based metrics fail to capture competition strength.

A model may output a factual answer while assigning substantial probability mass to counterfactual alternatives.

Therefore probabilistic evaluation is required.

A.4.2 Competition Metric

The competition score is defined as:

$$\Delta = \log p(A_{fact}) - \log p(A_{cf})$$

where:

- A_{fact} denotes the factual answer.
- A_{cf} denotes the counterfactual answer.

Interpretation:

- $\Delta > 0$: factual dominance.
- $\Delta < 0$: contextual dominance.
- $\Delta = 0$: balanced competition.

A.4.3 Example Calculation

Suppose:

$$\log p(A_{fact}) = -1.25$$

$$\log p(A_{cf}) = -3.75$$

Then:

$$\Delta = (-1.25) - (-3.75) = 2.50$$

indicating strong factual preference.

A.5 Appendix D: Bootstrap Statistical Analysis

A.5.1 Purpose

Bootstrap estimation is used to quantify uncertainty in competition scores.

A.5.2 Procedure

For each experiment:

1. Sample dataset with replacement.
2. Compute mean Δ.
3. Repeat 1000 times.
4. Estimate confidence intervals.

A.5.3 Confidence Interval Computation

Given bootstrap estimates:

$$\Delta_1, \Delta_2, \ldots, \Delta_n$$

The 95% confidence interval is:

$$[\Delta_{2.5},\ \Delta_{97.5}]$$

A.6 Appendix E: Model Configurations

A.6.1 GPT-2 Small

Table A.2: GPT-2 Small Configuration

Parameter	Value
Parameters	124M
Layers	12
Heads	12
Hidden Size	768
Vocabulary Size	50,257

A.6.2 GPT-2 Medium

Table A.3: GPT-2 Medium Configuration

Parameter	Value
Parameters	355M
Layers	24
Heads	16

A.6.3 GPT-2 Large

Table A.4: GPT-2 Large Configuration

Parameter	Value
Parameters	774M
Layers	36
Heads	20

A.6.4 Qwen2.5-3B

Table A.5: Qwen2.5-3B Configuration

Parameter	Value
Parameters	3B
Architecture	Decoder-only
Training Type	Pretrained

A.6.5 Qwen2.5-3B-Instruct

Table A.6: Qwen2.5-3B-Instruct Configuration

Parameter	Value
Parameters	3B
Architecture	Decoder-only
Training Type	Instruction Tuned

A.7 Appendix F: Hyperparameter Settings

Table A.7: Evaluation Hyperparameters

Parameter	Value
Temperature	0
Top-p	1.0
Top-k	Disabled
Max Tokens	10
Seed	Fixed
Batch Size	1

A.8 Appendix G: Attention Head Ablation Implementation

A.8.1 Overview

Each attention head is independently ablated using forward hooks.

Pseudo-code:

Algorithm 2 Single Head Ablation

Layer Attention Head

Register Forward Hook
Zero Head Output
Run Inference
Measure Δ
Remove Hook

A.8.2 Ablation Metric

For each head:

$$\Delta_{change} = \Delta_{ablated} - \Delta_{baseline}$$

Large absolute values indicate stronger causal influence.

A.9 Appendix H: Premise Word Categories

A.9.1 Definitional

- redefine
- define
- call
- name
- characterize

A.9.2 Verification

- verify
- validate
- fact check
- confirm
- cross-check

A.9.3 Evaluation

- assess
- review
- evaluate
- analyze
- examine

A.9.4 Hypothetical

- suppose
- imagine
- assume
- consider
- pretend

A.9.5 Reporting

- state
- note
- report
- according to
- describe

A.10 Appendix I: Experimental Workflow

Figure A.1 summarizes the complete experimental workflow.

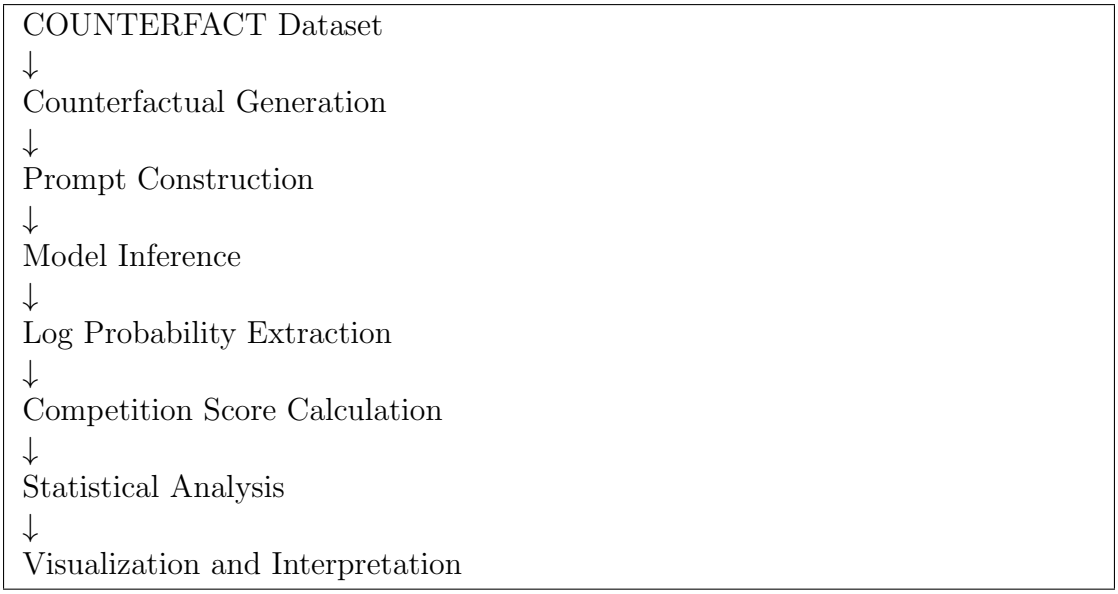


Figure A.1: Overall experimental pipeline.

A.11 Appendix J: Reproducibility Checklist

To reproduce the experiments:

1. Download dataset.
2. Install required dependencies.
3. Load pretrained models.
4. Generate prompts.
5. Run inference.
6. Extract token probabilities.
7. Compute competition scores.
8. Generate statistical summaries.
9. Produce visualizations.

A.12 Appendix K: Software Environment

Table A.8: Software Stack

Software	Version
Python	3.10+
PyTorch	2.x
Transformers	4.x
NumPy	Latest
Pandas	Latest
Matplotlib	Latest
SciPy	Latest

A.13 Appendix L: Repository Structure

```
project/
|
+-- datasets/
+-- notebooks/
|   +-- Objective_I/
|   +-- Objective_II/
|   +-- Objective_III/
|   +-- Ablation/
|
+-- results/
+-- figures/
+-- scripts/
|
+-- README.md
+-- requirements.txt
+-- environment.yml
```

A.14 Appendix M: Additional Notes

Several observations emerged during experimentation:

1. Context-memory competition is robust across datasets.
2. Semantic effects persist under multiple prompt formulations.
3. Structural effects remain observable even after filtering.
4. Scaling consistently increases contextual adaptability.
5. Alignment produces stronger regime-switching behavior.
6. Competition appears distributed across attention heads.

These observations further support the central conclusions of the thesis.

A.15 Appendix Summary

This appendix provided supplementary information regarding dataset construction, experimental design, model configurations, statistical analysis, reproducibility procedures, implementation details, and supporting materials.

Together with the results presented in the main chapters, these appendices ensure that the context-memory competition framework developed in this thesis can be independently reproduced, validated, and extended by future researchers.

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